

Chemoproteomics validates selective targeting of *Plasmodium* M1 alanyl aminopeptidase as a cross-species strategy to treat malaria

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

[About eLife's process](#)

Reviewed preprint version 1

February 2, 2024 (this version)

Sent for peer review

October 27, 2023

Posted to preprint server

August 17, 2023

Darren Creek , Carlo Giannangelo, Matthew Challis, Ghizal Siddiqui, Rebecca Edgar, Tess Malcolm, Chaille Webb, Nyssa Drinkwater, Natalie Vinh, Christopher MacRaild, Natalie Counihan, Sandra Duffy, Sergio Wittlin, Shane Devine, Vicky Avery, Tania de Koning-Ward, Peter Scammells, Sheena McGowan

Monash University • Deakin University • Griffith University • University of Basel • Walter and Eliza Hall Institute • Monash Institute of Pharmaceutical Sciences

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

New antimalarial drug candidates that act via novel mechanisms are urgently needed to combat malaria drug resistance. Here, we describe the multi-omic chemical validation of *Plasmodium* M1 alanyl metalloaminopeptidase as an attractive drug target using the selective inhibitor, MIPS2673. MIPS2673 demonstrated potent inhibition of recombinant *Plasmodium falciparum* (PfA-M1) and *Plasmodium vivax* (Pv-M1) M1 metalloaminopeptidases, with selectivity over other *Plasmodium* and human aminopeptidases, and displayed excellent *in vitro* antimalarial activity with no significant host cytotoxicity. Orthogonal label-free chemoproteomic methods based on thermal stability and limited proteolysis of whole parasite lysates revealed that MIPS2673 solely targets PfA-M1 in parasites, with limited proteolysis also enabling estimation of the binding site on PfA-M1 to within ~5 Å of that determined by X-ray crystallography. Finally, functional investigation by untargeted metabolomics demonstrated that MIPS2673 inhibits the key role of PfA-M1 in haemoglobin digestion. Combined, our unbiased multi-omic target deconvolution strategies confirmed the on-target activity of MIPS2673, and validated selective inhibition of M1 alanyl metalloaminopeptidase as a promising multi-stage and cross-species antimalarial strategy.

eLife assessment

The manuscript constitutes an **important** contribution to antimalarial drug discovery, employing diverse systems biology methodologies; with a focus on an improved M1 metalloprotease inhibitor, the study provides **convincing** evidence of the utility of chemoproteomics in elucidating the preferential targeting of PfA-M1. Additionally, metabolomic analysis effectively documents specific alterations in the final steps of hemoglobin breakdown. These findings underscore the potential of the developed methodology, not only in understanding PfA-M1 targeting but also in its broader applicability to diverse malarial proteins or pathways. Revisions are needed to further enhance overall clarity and detail the scope of these implications.

Introduction

Malaria is a devastating parasitic disease that causes major health, economic and societal impacts to almost half of the world's population that live in malaria-endemic countries. In 2021 there were an estimated 247 million cases of malaria globally and 619 000 associated deaths.¹ An estimated 95% of malaria deaths are attributed to *P. falciparum*, the species responsible for the most severe and lethal form of the disease.¹ *P. vivax* is also responsible for significant morbidity, and is the most geographically widespread malaria parasite.² Although the global incidence rate of malaria has declined over the last decade, the number of people worldwide suffering and dying from malaria is now increasing and parasite resistance has emerged to all current antimalarial drug classes, including the front-line artemisinin therapies.^{1,3} Alarmingly, artemisinin-based combination therapies fail to cure infections in up to 50% of patients in some regions of Asia⁴ and delayed parasite clearance rates following artemisinin-based therapy (synonymous with decreased parasite susceptibility to artemisinin) has now been detected in Africa,⁵ where most malaria deaths occur. This, combined with the lack of mechanistic diversity in emerging antimalarials in development, means there is an urgent need for new drug classes with novel mechanisms of action that bypass resistance and treat both *P. falciparum* and *P. vivax* infections.

Haemoglobin digestion is an essential metabolic process and a point of key vulnerability during *Plasmodium* parasite intraerythrocytic development. The parasite digests up to 75% of host cell haemoglobin to provide amino acids for protein synthesis and this process additionally serves non-anabolic functions, such as maintaining osmotic stability of the infected red blood cell.^{6–9} *Plasmodium* parasites employ the interplay of numerous proteases from different classes to digest haemoglobin into its constituent amino acids.¹⁰ *P. falciparum* M1 alanyl aminopeptidase (*PfA-M1*) and M17 leucyl aminopeptidase (*PfA-M17*) are two zinc-dependent metalloaminopeptidases involved in the terminal stage of haemoglobin digestion.^{11,12} Both *PfA-M1* and *PfA-M17* are thought to be essential for parasite survival and are promising targets for antimalarial therapy *in vitro* and *in vivo*.^{11,25} However, attempts at genetic manipulation of *PfA-M1* have been unsuccessful to date, which has precluded formal genetic validation of *PfA-M1* as a drug target. Bestatin-based inhibitors of *PfA-M1* were previously shown to prevent proteolysis of haemoglobin peptides, leading to a swollen digestive vacuole and stalled growth at the trophozoite stage.¹⁵ Loss of *PfA-M17* function through chemical inhibition or conditional knockdown also results in stalled growth at the trophozoite stage but does not induce digestive vacuole swelling.¹¹ Instead, specific loss of *PfA-M17* function results in the presence of multiple membrane bound digestive vacuoles per parasite. We have previously used rational drug design to optimise dual M1 and M17 inhibitors with nanomolar potency across multiple *Plasmodium* species.^{20,21,23,25} More recently, we reported a selective *PfA-M17* inhibitor with potent antimalarial activity.¹¹

Like *PfA-M17*, *PfA-M1* is an attractive target for novel antimalarial drug development. The *PfA-M1* enzyme is a monomeric protein consisting of four domains and is widely believed to function in the cytosol, although localisation to other compartments has been reported.^{12,15,26–28} It has a broad substrate specificity that includes most basic and hydrophobic amino acids, but preferentially cleaves leucine and methionine.^{26,29} *PfA-M1* uses a zinc-dependent mechanism to catalyse the release of single amino acids from the N-terminus of short peptides. All reported inhibitors of *PfA-M1* target the enzyme active site and act to chelate the catalytic zinc ion.^{13,15,19,21,29–33} Obtaining selectivity for *PfA-M1* over *PfA-M17* is challenging as the enzymes share similar architecture within the active sites and a catalytic mechanism involving zinc ions.^{30,34} As such, very few compounds that selectively inhibit *PfA-M1* over *PfA-M17* have been reported^{15,18,31} and unbiased identification of the target in parasites is critical to confirm specificity.

Identifying the molecular targets of compounds is a significant challenge for antimalarial drug development.³⁵ One of the most successful approaches to target deconvolution has been *in vitro* evolution and whole genome analysis (IVIEWGA).^{37–41} However, IVIEWGA can also point to genes encoding general drug-resistance mechanisms rather than the cellular target, and a major goal of new drug development for malaria is the identification of compounds that fail to yield resistant parasites *in vitro* (irresistible compounds), as these compounds are less likely to result in clinical resistance.⁴² Chemoproteomics methods based on mass spectrometry offer alternative routes to validate the molecular target(s) of compounds in parasites. One strategy relies on the use of chemical probes to enrich and identify protein targets.⁴³ However, this method requires chemical modification of the drug, which could alter its pharmacological properties. New thermal stability^{44,45} and limited proteolysis⁴⁶ techniques are two examples of unbiased chemoproteomics methods that provide direct evidence of on-target engagement on a proteome-wide scale, without the need for modification of the drug. These methods rely on the concepts of thermal and proteolytic stabilisation of the protein target due to ligand binding and leverage recent advances in mass spectrometry to detect drug-specific changes to protein or peptide abundance within complex proteomes. Thermal proteome profiling was recently applied in *P. falciparum* to successfully identify the targets of several antimalarial compounds.^{45,47} Whereas limited proteolysis proteomics has not previously been reported to identify drug targets in *P. falciparum*, this approach has successfully mapped protein-small molecule interactions in other complex eukaryotic proteomes.⁴⁶

Here, we report a multi-stage M1 selective aminopeptidase inhibitor with a hydroxamic acid zinc binding group²⁰ (MIPS2673), which targets both the *P. falciparum* (PfA-M1) and *P. vivax* (Pv-M1) M1 homologues and demonstrates antimalarial activity against a panel of drug resistant *P. falciparum* strains. We developed orthogonal chemoproteomics approaches based on thermal stability and limited proteolysis, combined with metabolomics analysis, to validate the on-target activity of MIPS2673 in parasites. Our comparatively simple and inexpensive chemoproteomics methods for identifying antimalarial drug targets utilise an efficient data-independent acquisition mass spectrometry (DIA-MS) approach that allows comprehensive and reproducible proteome coverage. Together, our studies confirm *Plasmodium* M1 alanyl aminopeptidase as the sole target of MIPS2673 and validate selective inhibition of this enzyme as a multi-stage and cross-species antimalarial strategy.

Results

Design and synthesis of a cross-species hydroxamic acid-based inhibitor selective for *Plasmodium* M1 alanyl aminopeptidase

We previously reported a series of PfA-M1 and M17 aminopeptidase inhibitors that possessed a hydroxamic acid zinc binding group to coordinate catalytic zinc ion(s) and a variety of hydrophobic groups to probe the S1' binding pocket of these enzymes.²¹ Whilst a number of these modifications successfully improved inhibitory potency, their incorporation reduced polarity and aqueous solubility. One such example is the pivaloyl group present in compound **1** (Fig. 1A).^{20,21} In order to balance these opposing factors, one of the methyl groups of the pivaloyl was replaced by an alcohol to increase polarity whilst maintaining the capacity to engage the S1'-pocket via hydrophobic interactions (Fig. 1A). This modification resulted in MIPS2673 (compound **4**), which has a significantly lower cLogP (1.45 vs 2.23 for **1**) and favourable solubility (i.e. the kinetic solubility of MIPS2673 was estimated to be in the 50–100 µg/mL range using nephelometry vs 12.5–25 µg/mL for **1**). MIPS2673 (**4**) was synthesized from methyl 2-amino-2-(4-bromophenyl)acetate (**2**) in three synthetic steps as shown in Fig. 1. Amide formation was achieved via an EDCI-mediated coupling and the trifluorophenyl moiety was installed using a

Suzuki-Miyaura coupling to form the synthetic intermediate **3**. Finally, the methyl ester was converted to the corresponding hydroxamic acid using 5 M methanolic KOH and $\text{NH}_2\text{OH}\cdot\text{HCl}$ to afford MIPS2673 (**4**).

We determined the inhibitory activity of MIPS2673 against M1 and M17 enzymes from both *P. falciparum* and *P. vivax* (**Fig. 1B**). The binding affinities (K_i) of MIPS2673 toward purified, recombinant PfA-M1 shows the compound to be a potent inhibitor ($K_i = 211 \pm 11$ nM), with > 4-fold selectivity over PfA-M17 ($K_i = 921 \pm 69$ nM). MIPS2673 was significantly more potent against Pv-M1 ($K_i = 6.4 \pm 0.5$ nM) with > 155-fold selectivity over Pv-M17 ($K_i > 1000$ nM). We confirmed that MIPS2673 is non-inhibitory against the PfA-M18 ($K_i > 500$ μM) and PfAPP ($K_i > 40$ μM) recombinant metalloaminopeptidases (data not shown).

In contrast to the M1 selectivity demonstrated by MIPS2673, the inhibitor **1** was 2-fold more selective for PfA-M17 vs PfA-M1.²¹ When bound to PfA-M1, the position of the biaryl of MIPS2673 compares with that of our previous inhibitors.^{20,21} The moiety makes similar interactions with Met1034 (hydrophobic) and Glu319 (carbonyl- π) (**Fig. 1B**). The fluoro-substituents sit deep in the S1 pocket and form the same intricate network of water-mediated hydrogen-bonds as **1**, in which the fluorine atoms act as H-bond acceptors. The substitution on the *tert*-butyl, replacing a methyl with an alcohol, is difficult to definitively position in the structure due to the rotation around the C-O3 bond. It is possible that this alcohol could form a long H-bond to Arg-489, which should provide some improvement in position and potency. The Arg-489 side-chain shows two orientations in the electron density, suggesting that some rotation of the alcohol is occurring (**Fig. 1B**). Inspection of the structure of PfA-M17 with **1** (PDB ID 4ZY2)²⁰ does not reveal why the compound is selective. The *tert*-butyl of **1** is sitting in an open pocket and the introduction of an alcohol is unlikely to induce any clashes, steric or electrostatic. This suggests that the reason for the selectivity between the two enzymes results from the ability (or inability) to access the interior cavity of the PfA-M17 hexamer that houses the active sites.

To investigate the potential for off-target effects, we examined the selectivity of MIPS2673 over several human M1 homologs, including leukotriene A-4 hydrolase (LTA4H), endoplasmic reticulum aminopeptidase 1 (ERAP1) and endoplasmic reticulum aminopeptidase 2 (ERAP2). MIPS2673 showed minimal inhibition of ERAP1 and ERAP2 at concentrations below 500 μM (Table S1), whereas MIPS2673 caused > 50% inhibition of LTA4H at concentrations above 10 μM (Table S1). We also tested MIPS2673 against the HEK293 cell line to further predict potential human cytotoxicity. No cytotoxicity to HEK293 cells was observed up to a concentration of 40 μM . As a comparison, the current antimalarials chloroquine and dihydroartemisinin inhibit HEK293 proliferation by 67% and 29%, respectively, at 40 μM . At 120 μM , MIPS2673 inhibited cellular proliferation by 89% compared to the untreated control (Table S2).

In vitro antiparasitic activity of MIPS2673

Having synthesised a *Plasmodium* specific M1 aminopeptidase inhibitor, we next explored the activity of MIPS2673 against *in vitro* blood-stage cultures of *P. falciparum*. We found that MIPS2673 has sub micromolar activity against the laboratory reference strain, 3D7 (72 h $\text{EC}_{50} = 324$ nM (250–470 CI)) (**Fig. 1C**), as well as against a panel of drug resistant *P. falciparum* lines (Table S3). Stage specificity assays were performed to identify which stage of asexual development MIPS2673 is most active against. Synchronised ring or trophozoite cultures were exposed to MIPS2673 for 24 h or 48 h at 10x EC_{50} , then stringently washed to remove the compound. Parasite growth was assessed after 48 h compared to a vehicle-treated control. This confirmed MIPS2673 is most potent against trophozoite stage parasites (**Fig. 1D**), corresponding to the period of peak PfA-M1 expression²⁶ and activity.⁴⁸ As expected, no stage-dependent difference in activity was observed in parasites exposed to MIPS2673 for 48 h, as this duration incorporates the full asexual erythrocytic cycle. The control compound, artesunate, was equipotent across all conditions, consistent with the known activity and kinetic profiles for artemisinins.⁴⁹

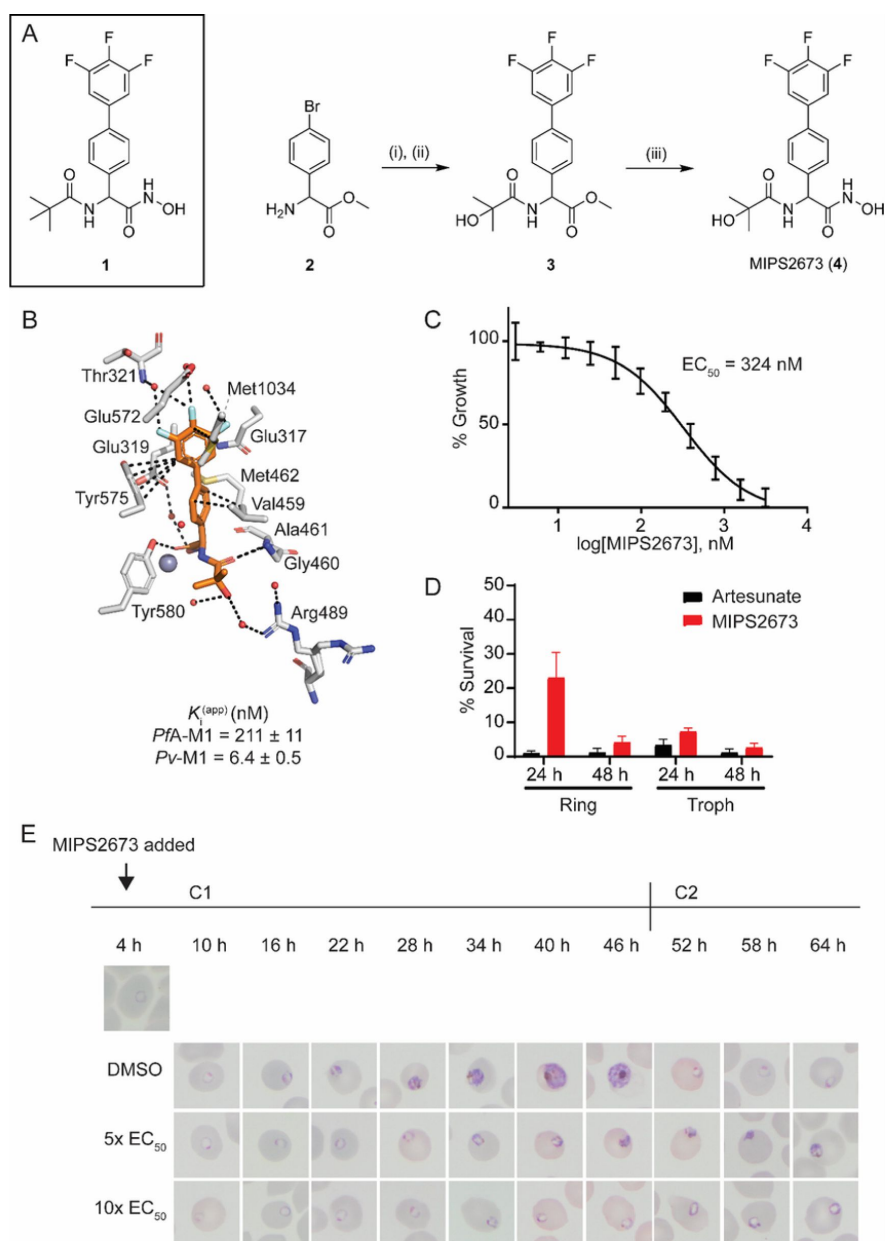


Figure 1

Synthesis and antimalarial activity of the M1-selective aminopeptidase inhibitor, MIPS2673.

(A) Synthesis of MIPS2673 (4). *Reagents and conditions:* (i) RCOOH, EDCI, DMAP, CH_2Cl_2 ; (ii) 3,4,5-trifluorophenylboronic acid, $Pd(PPh_3)_2Cl_2$, Na_2CO_3 , THF; (iii) $NH_2OH \cdot HCl$, KOH. (B) Binding mode of MIPS2673 bound to *PfA*-M1. Interactions between MIPS2673 (orange sticks) and *PfA*-M1 (grey) shown as black dashed lines. Inhibition constant (K_i) for MIPS2673 toward recombinant, purified *PfA*-M1 and *Pv*-M1. (C) Sensitivity to MIPS2673 exposure (72 h) for *P. falciparum* 3D7 parasites. Parasite growth was determined relative to a no-drug control by SYBR Green I assay. The EC_{50} value was calculated from 4 biological replicates performed in technical triplicate and the data plotted as the mean $\pm$ standard error of the mean. (D) Stage-dependent activity of MIPS2673 in *P. falciparum* (3D7). Synchronised ring- or trophozoite-stage cultures were incubated for either 24 or 48 h with compound at 10x EC_{50} , before the drug was washed off and parasites allowed to grow for a further 48 hours. Growth was determined via Sybr Green I assay and compared to vehicle (DMSO)-treated controls. Shown is the mean $\pm$ standard deviation ($n=4$). (E) MIPS2673-induced morphology and growth effects for *P. falciparum* 3D7 parasites. Synchronised cultures at 4 h post-invasion were treated over two cycles (C1, cycle 1; C2, cycle 2) with either 5x or 10x EC_{50} or DMSO at the concentration present in the 10x EC_{50} treatment. Representative Giemsa-stained smears from 2 biological replicates show parasites stalling and dying at early trophozoite stage (5x EC_{50}) or stalling at ring stage (10x EC_{50}).

To visually characterise the effects of MIPS2673 on parasites, we treated synchronised 3D7 cultures at the early ring stage and monitored their development by light microscopy evaluation of Giemsa-stained thin blood smears (**Fig. 1E** [↗](#)). We found that MIPS2673 caused significant growth retardation when compared to the untreated control and parasite growth stalled at early trophozoite (22–28 h trophozoites) and late ring (~ 22 h rings) stages at 5x and 10x EC₅₀, respectively. The presence of haemozoin in both treatment groups indicated that the initial catabolism of haemoglobin is unaffected by MIPS2673.

Having confirmed activity against asexual stages of *P. falciparum*, we next determined whether MIPS2673 kills the transmissible sexual forms of the parasite, known as gametocytes. We found that MIPS2673 has transmission-blocking activity and is most potent against early gametocytes (stages I–III), compared to late (stages IV–V) and mature (stage V) stages (**Table S4** [↗](#)), but is less potent than the artemisinin derivative, artesunate.

Identification of molecular targets for MIPS2673 in *P. falciparum* by thermal stability proteomics

Our biochemical studies showing binding and inhibition of recombinant *Plasmodium* metalloaminopeptidases do not rule out the possibility that the anti-parasitic activity of MIPS2673 is due to non-specific binding to other parasite metalloproteins or off-target proteins. Genetic target validation was not possible as previous attempts to mutate *PfA-M1* were not successful,^{12,50} and MIPS2673-resistant parasites were not available. Therefore, we developed orthogonal mass spectrometry-based chemoproteomics approaches to validate that MIPS2673 is on-target and M1-selective in the complex parasite environment (**Fig. 2** [↗](#)). We initially developed a streamlined thermal stability proteomics workflow that combined traditional thermal proteome profiling methods with an optimised data-independent acquisition (DIA)-LC-MS/MS approach.^{51,52}

To identify the binding target/s of MIPS2673 in *P. falciparum* asexual blood stages in a proteome-wide manner, we used native parasite lysates, as direct drug-protein interactions are more selectively identified in cellular lysates, rather than live cells that are susceptible to downstream effects of drug action. The parasite lysates were exposed to 1 μ M or 4 μ M of MIPS2673 or vehicle (DMSO control) for 3 minutes prior to heating at 60°C, a temperature that should allow detection of most drug-induced protein stabilisation events in an untargeted manner with wide proteome coverage.^{45,53} After the thermal challenge, the soluble (non-denatured) protein fraction was isolated by ultracentrifugation, digested overnight with trypsin and the peptide mixture analysed directly using global DIA-LC-MS/MS. Proteins detected with significantly higher abundance in treated relative to control samples reflect thermal stabilisation of the target due to ligand binding. To minimise identification of false positive hits, proteins reproducibly stabilised ($p < 0.05$ and fold-change ≥ 1.2 compared to the untreated control) across multiple drug concentrations and experiments were considered drug interacting proteins. Among 1632 proteins reproducibly detected with a minimum of two peptides across two independent experiments (each with at least three independent incubations of MIPS2673 or vehicle with protein lysate), five proteins were consistently stabilised at both drug concentrations compared to DMSO (**Fig. 3A** [↗](#), **Data S1** [↗](#) and **Data S2** [↗](#)). Of these, *PfA-M1* (PF3D7_1311800) was one of the most significantly stabilised proteins ($p < 0.01$ at 1 μ M). *PfA-M1* exhibited concentration-dependent stabilisation in the presence of MIPS2673, with an average stabilisation of 1.8-fold and 2.8-fold at 1 μ M and 4 μ M, respectively, relative to the untreated control (**Fig. 3B** [↗](#)). The four other proteins consistently stabilised were not metalloproteins and included two conserved *Plasmodium* proteins with unknown functions (PF3D7_1026000 and PF3D7_0604300), a putative AP2 domain transcription factor (PF3D7_1239200) and a human protein, Ras-related Rab-39A (Q14964) (**Fig. 3B** [↗](#)). These proteins could represent possible off-target binding. The *PfA-M17* protein was not stabilised by MIPS2673 at either drug concentration after a 60°C thermal challenge (**Fig. S1** [↗](#)). We previously showed *PfA-M17* is stabilised at this temperature in parasite lysates treated with a selective *PfA-M17*

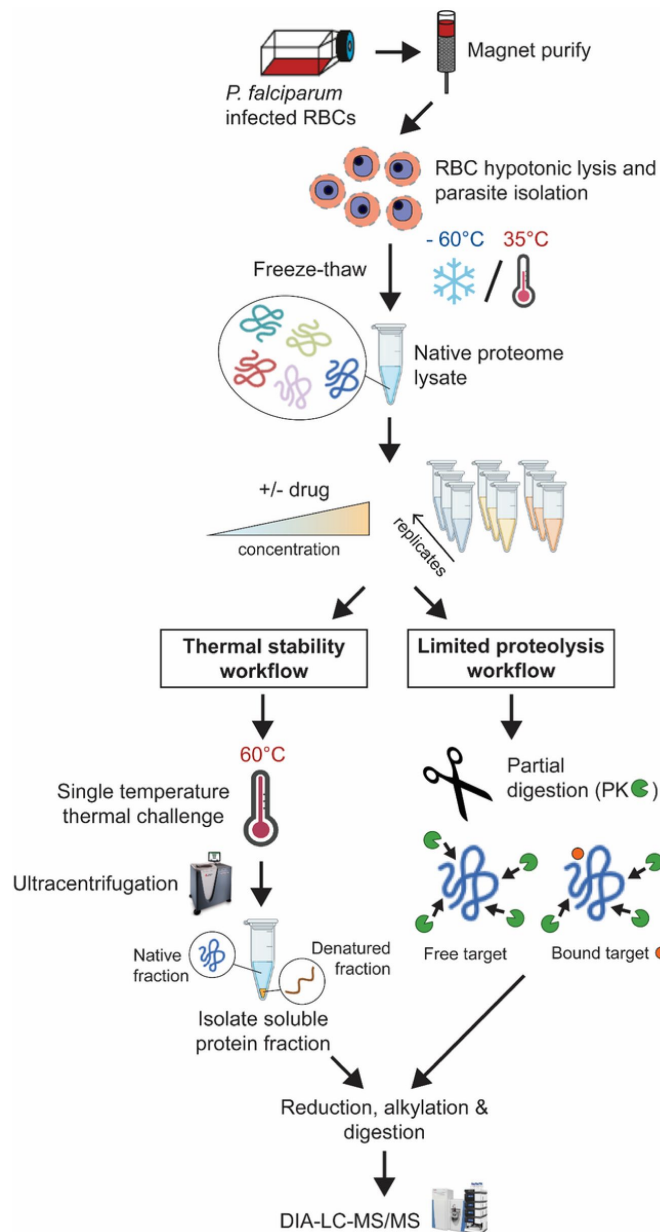


Figure 2

Experimental thermal stability and limited proteolysis workflows for unbiased drug target identification in *P. falciparum* lysates.

Parasites are isolated from the host red blood cell by hypotonic lysis. Proteins are extracted from cells under native conditions by multiple freeze-thaw cycles. The lysate is pooled and distributed to achieve multiple independent incubations per reaction condition for use in either thermal stability or limited proteolysis workflows. For thermal stability proteomics studies, the parasite lysate is subjected to treatment with vehicle (untreated control) or drug at two concentrations, followed by thermal challenge at 60 °C. The soluble (non-denatured) protein fraction is isolated by ultracentrifugation and digested overnight with trypsin. The resulting peptide mixture is analysed by DIA-LC-MS/MS against an in-house generated spectral library to determine drug-induced thermal stabilisation. For limited proteolysis studies, ligand-protein interactions are monitored at multiple drug concentrations following double protease digestion. An initial pulse proteolysis with the broad specificity protease, proteinase K (PK), captures local ligand-induced structural alterations of proteins that become differentially susceptible to protease cleavage upon drug binding. Samples then undergo complete digestion under reducing conditions with trypsin/LysC. Differential proteolytic patterns produced upon ligand binding are detected by DIA-LC-MS/MS and analysed against a limited proteolysis spectral library.

inhibitor.¹¹ Overall, our unbiased thermal stabilisation proteomics approach confirmed MIPS2673 selectively targets the M1 aminopeptidase over M17 and does not bind indiscriminately to parasite metalloproteins.

Identification of MIPS2673 targets in *P. falciparum* by limited proteolysis coupled mass spectrometry

We also developed an efficient protocol for limited proteolysis-based studies of *P. falciparum* and applied it for the unbiased identification of targets of MIPS2673 (Fig. 2). Native *P. falciparum* lysates were treated for 10 minutes with different concentrations of MIPS2673 (1 μ M or 10 μ M in experiment one and 0.1 μ M, 1 μ M or 10 μ M in experiment two) or vehicle in at least four independent incubations per experiment. Proteome extracts were then subjected to double protease digestion. An initial limited proteolysis with proteinase K for 4 minutes captured local structural alterations of proteins that become differentially susceptible to protease cleavage upon drug binding. Secondly, samples undergo complete digestion under reducing conditions overnight with trypsin/LysC to make peptides amenable for global proteomics analysis. Differentially abundant proteolytic peptides between MIPS2673 and vehicle-treated samples were then identified on a global scale with DIA-LC-MS/MS. The proteolytic peptide patterns of drug targets should be altered in the treated samples as compound binding prevents protein cleavage by proteinase K, resulting in decreased abundance of peptides with non-tryptic ends or an increase in concentration of the associated fully tryptic peptide (Fig. 4A). We quantified 26,611 peptides from 2,153 proteins in experiment one and 16,662 peptides from 1,989 proteins in experiment two (Data S3). Each dataset was initially analysed for differentially abundant peptides between each MIPS2673 concentration and vehicle by filtering based upon relative peptide abundance (absolute fold-change > 1.5), statistical significance ($q < 0.01$) and proteolytic peptide pattern (i.e. increased fully tryptic or decreased half tryptic). In experiment one and experiment two respectively, approximately 3% of peptides from 379 proteins and 0.6% of peptides from 69 proteins, met these thresholds at each drug concentration and were considered significant (Fig. 4B). After prioritising targets based upon the number of significant peptides detected per protein (Fisher exact test, Bonferroni corrected $p < 0.05$), we identified seven putative drug targets in experiment one and only one putative target in experiment two (Fig. 4C and Data S4). Interestingly, among the seven possible target proteins identified by limited proteolysis in experiment one, five were metalloproteins. These included PfA-M1, three other aminopeptidases - M17 leucyl aminopeptidase, aminopeptidase P (PfAPP) and M18 aspartyl aminopeptidase (PfA-M18) - and adenosine deaminase (PfADA) (Fig. 4C). The expected target, PfA-M1 aminopeptidase, was the only protein reproducibly identified as the MIPS2673 target across both independent limited proteolysis experiments. Indeed, the consistent identification of PfA-M1 as the only significant protein in both LiP-MS and thermal stability proteomics methods strongly supports PfA-M1 being the primary target of MIPS2673.

Features of structurally significant Pf A-M1 LiP-MS peptides

The identification of drug targets with LiP-MS typically identifies structurally perturbed peptides located in very close proximity to the ligand binding site.^{46,54} Among the 108 PfA-M1 peptides reproducibly detected across both LiP-MS experiments we identified nine structurally significant LiP-MS peptides ($q < 0.01$ and absolute fold-change > 1.5 at all drug concentrations in both experiments) (Fig. 5A) and mapped these to our PfA-M1 crystal structure with MIPS2673 bound. We measured the minimum distance between atoms of the significant peptides and those of MIPS2673 and found that structurally significant peptides identified with LiP-MS were frequently located in very close proximity to MIPS2673. The median minimum distance between the significant LiP-MS peptides and bound MIPS2673 was 6.5 Å, significantly less than the 13.9 Å median distance for all other detected PfA-M1 peptides (Fig. S2). Among the nine structurally significant LiP-MS peptides, four contained atoms within Van der Waals distance (< 4 Å) of

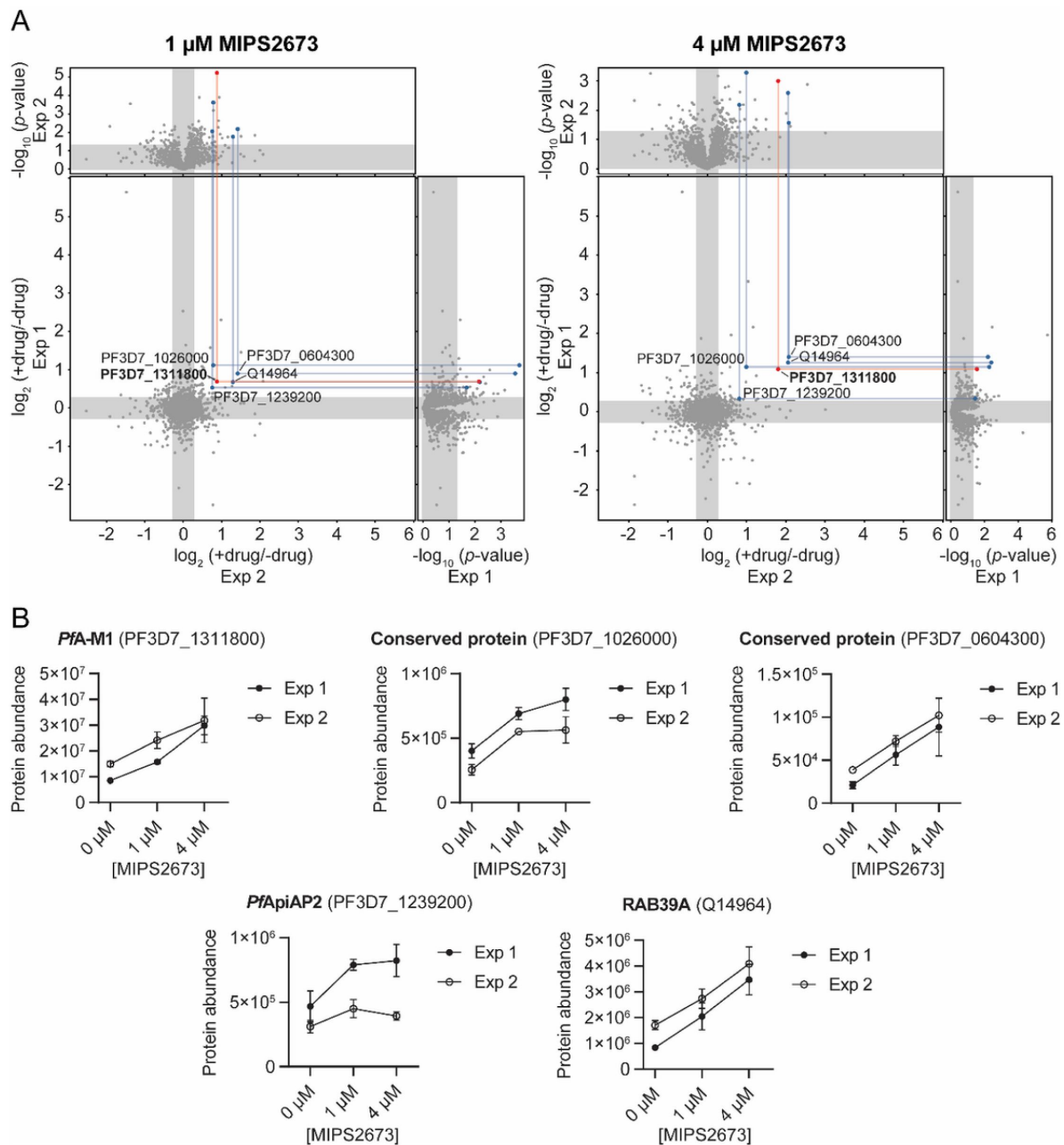


Figure 3

Thermal stabilisation of proteins by MIPS2673.

(A) Paired volcano plots of all proteins detected after a 60 °C thermal challenge across two experiments from *P. falciparum* lysates (at least n=3 of independent lysate incubations per condition and experiment) treated with 1 μM (left) and 4 μM (right) MIPS2673. Outside panels show the fold-change in protein abundance for treated relative to untreated samples as a function of statistical significance (p-value) for independent experiments. Each protein is represented with a single data point. Significance cut-offs for thermal stabilisation were $p < 0.05$ (Welch's t test) and fold-change > 1.2 (unshaded regions). The central panel plots protein fold-change from experiment one and two, with proteins in the upper right quadrant reproducibly stabilised by > 1.2 -fold. Proteins passing both fold-change and statistical cut-offs at both drug concentrations and in multiple independent experiments are highlighted. M1-family alanyl aminopeptidase (PF3D7_1311800) is in red and all other proteins passing cut-offs are in blue. (B) Individual profiles for the five proteins reproducibly stabilised by MIPS2673 after a 60 °C thermal challenge. Two independent experiments are shown. Data are mean protein abundances in the absence and presence of MIPS2673 ($\pm$ SD) from 4 replicates.

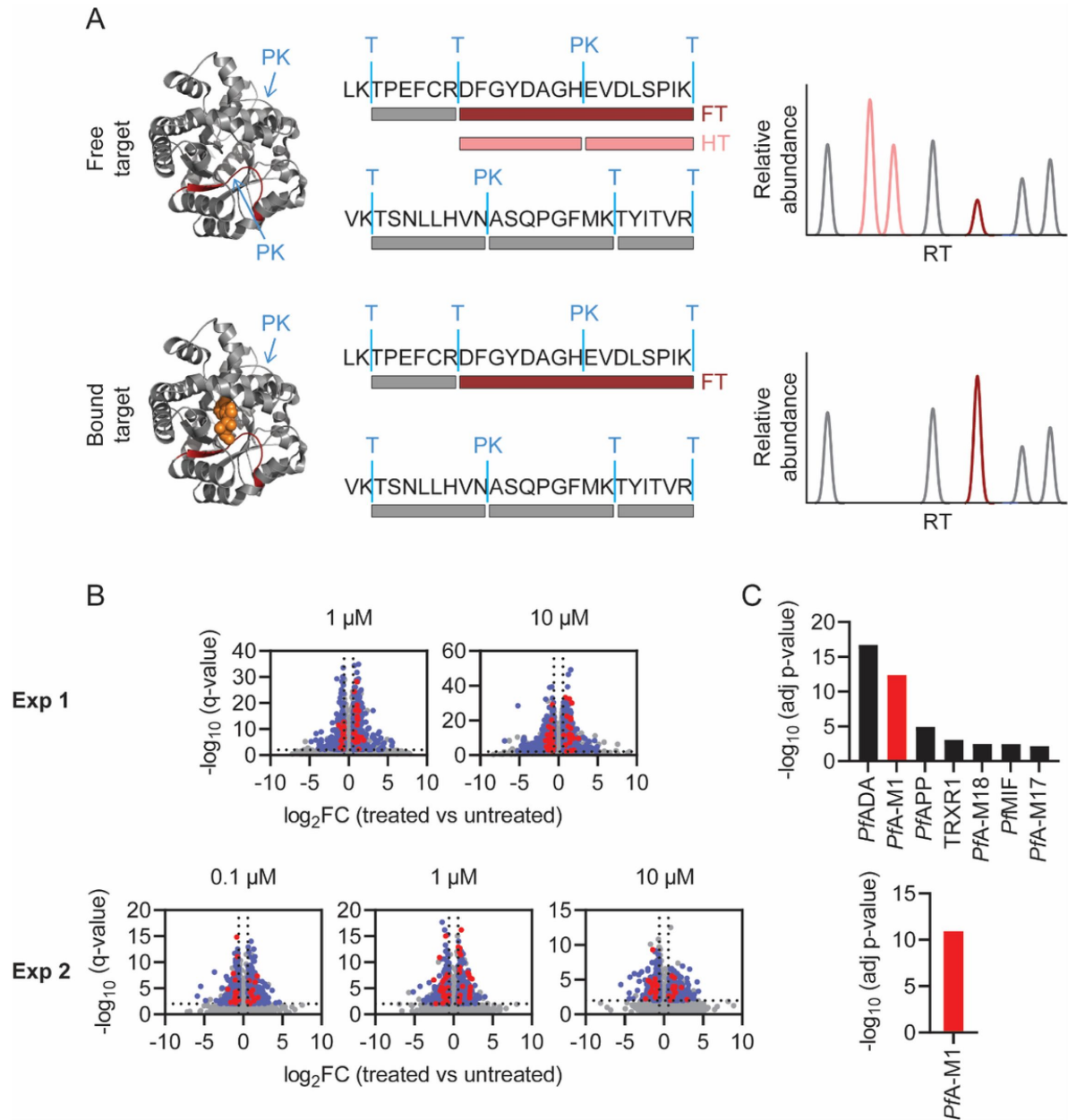


Figure 4

Identification of MIPS2673 target proteins using LiP-MS.

(A) Principle of LiP-MS analysis. Ligand binding to a protein alters the local proteolytic susceptibility and prevents protein cleavage by proteinase K. In the bound state this results in decreased abundance of peptides with non-tryptic ends (half tryptic, HT) shown in pink and/or an increase in concentration of the corresponding fully tryptic (FT) peptide, shown in red. Peptides unaffected by ligand binding (grey) are unchanged in abundance. (B) Volcano plots of two independent LiP-MS experiments (at least $n = 3$ independent lysate incubations per condition and experiment). Each data point represents a detected peptide. Fold-changes (FC) in peptide abundance in treated vs untreated samples are shown as a function of significance ($q < 0.01$) at the MIPS2673 concentrations shown. Peptides highlighted in blue represent LiP peptides that were significant across all concentrations tested within an experiment. Red peptides represent significant LiP peptides from PfA-M1. (C) Short-list of putative MIPS2673 targets identified in each LiP-MS experiment after prioritising targets based upon the number of significant peptides detected per protein. Significance was determined using a Fisher exact test and Bonferroni corrected $p < 0.05$. The expected target PfA-M1 aminopeptidase is shown in red.

MIPS2673. Based on these observations, we hypothesised that significantly dysregulated LiP-MS peptides could estimate the known MIPS2673 binding site on *PfA-M1*. A distance of 6.44 Å from MIPS2673 was used to define the binding cleft boundary⁵⁴ and included the *PfA-M1* residues known to interact with bound MIPS2673. Six of the nine significant LiP peptides overlapped with, or were within 4 Å of this binding cleft. The median distance between the atoms of significant LiP peptides and of cleft residues was 2.6 Å, compared to a median distance of 8.2 Å for all other *PfA-M1* peptides (Fig. 5B). Of the three significant LiP peptides located > 4 Å from the operational binding site, two are located at the C-terminal opening in domain IV that forms a channel leading towards the active site.³⁰ To understand if this type of data could approximate a ligand binding site correctly, we calculated the centre of mass of the atoms of the nine structurally significant LiP peptides and represented this as a geometric point within the *PfA-M1* structure and then compared this to the known binding site (Fig. 5C). The minimum distance between this point and MIPS2673 was 5.2 Å and the MIPS2673 centre of mass neighbourhood (residues within 6.44 Å of the centre of mass) overlapped with the binding site (Fig. 5C). This indicated that our LiP-MS approach provides a good approximation of the MIPS2673 binding site with its target, *PfA-M1*.

Untargeted metabolomics analysis of *P. falciparum* infected red blood cells treated with MIPS2673

To further determine the specificity of MIPS2673 for *PfA-M1* and whether it induces off-target effects, we performed untargeted metabolomics on infected red blood cells treated with 1 µM of MIPS2673 (3x EC₅₀ value) for 1 h and compared the profile to vehicle (DMSO) control (4–9 biological replicates). Heatmap analysis of relative abundances of all putative metabolites revealed that treatment with MIPS2673 disproportionally impacted peptide metabolism (Fig. S3). Of the 201 putative peptides identified, 97 were significantly dysregulated ($p < 0.05$). The majority of dysregulated peptides were significantly increased (fold-change > 1.5 and $p < 0.05$) in abundance in MIPS2673 treated cultures compared to DMSO control, indicative of aminopeptidase inhibition. Targeted analysis of the 97 increased peptides in drug treated cultures revealed that the majority were likely derived from haemoglobin (Hb) chains (α and β) (Fig. 6A). Mapping to Hb sequences was possible for 24 peptides using MS/MS spectra to confirm peptide sequences (Fig. 6B; green dots). For the remaining putative peptides identified by accurate mass, but for which MS/MS spectra could not be obtained, we assessed whether any peptide isomeric to the putative peptide could be mapped to Hb (Fig. 6B; orange dots for peptides mapping to Hb chains and blue dots for peptides that could not be mapped). Overall, ~80% of significantly dysregulated peptides could be mapped to one of the Hb chains, with nearly all increasing in abundance following treatment with MIPS2673 compared to DMSO control. To further validate that the majority of elevated peptides are likely to be Hb-derived, we repeated this same analysis for each of the ~4700 proteins identified in our recent comprehensive proteomic analysis of *P. falciparum*-infected red blood cells.⁵¹ We quantified the number of peptide matches to each protein and then divided by protein length to yield a normalized estimate of the similarity of each protein to our significantly dysregulated peptides. By this measure, Hb chains α , and β were the most highly matched protein compared to the remaining infected red blood cell proteome (Fig. 6C; red bars), indicating that MIPS2673 predominantly, but not exclusively, disrupts Hb digestion.

We have recently reported that an inhibitor of *PfA-M17* (MIPS2571) also predominantly disrupted metabolism of short haemoglobin-derived peptides, which was a metabolic signature consistent with genetic knockdown of *PfA-M17*.¹¹ However, a direct comparison of peptide perturbations induced by MIPS2673 and MIPS2571 (data from Edgar *et al*, 2022) showed clearly distinct peptide profiles. Whilst many peptides accumulated with both inhibitors, the extent of peptide accumulation differed, and a subset of short basic peptides (containing Lys or Arg) were elevated following exposure to MIPS2673, but not the *PfA-M17* inhibitor (Fig. 6A). Furthermore, the peptide changes observed following treatment with MIPS2673 differ substantially from the peptide changes observed following treatment of *P. falciparum*-infected red blood cells with other

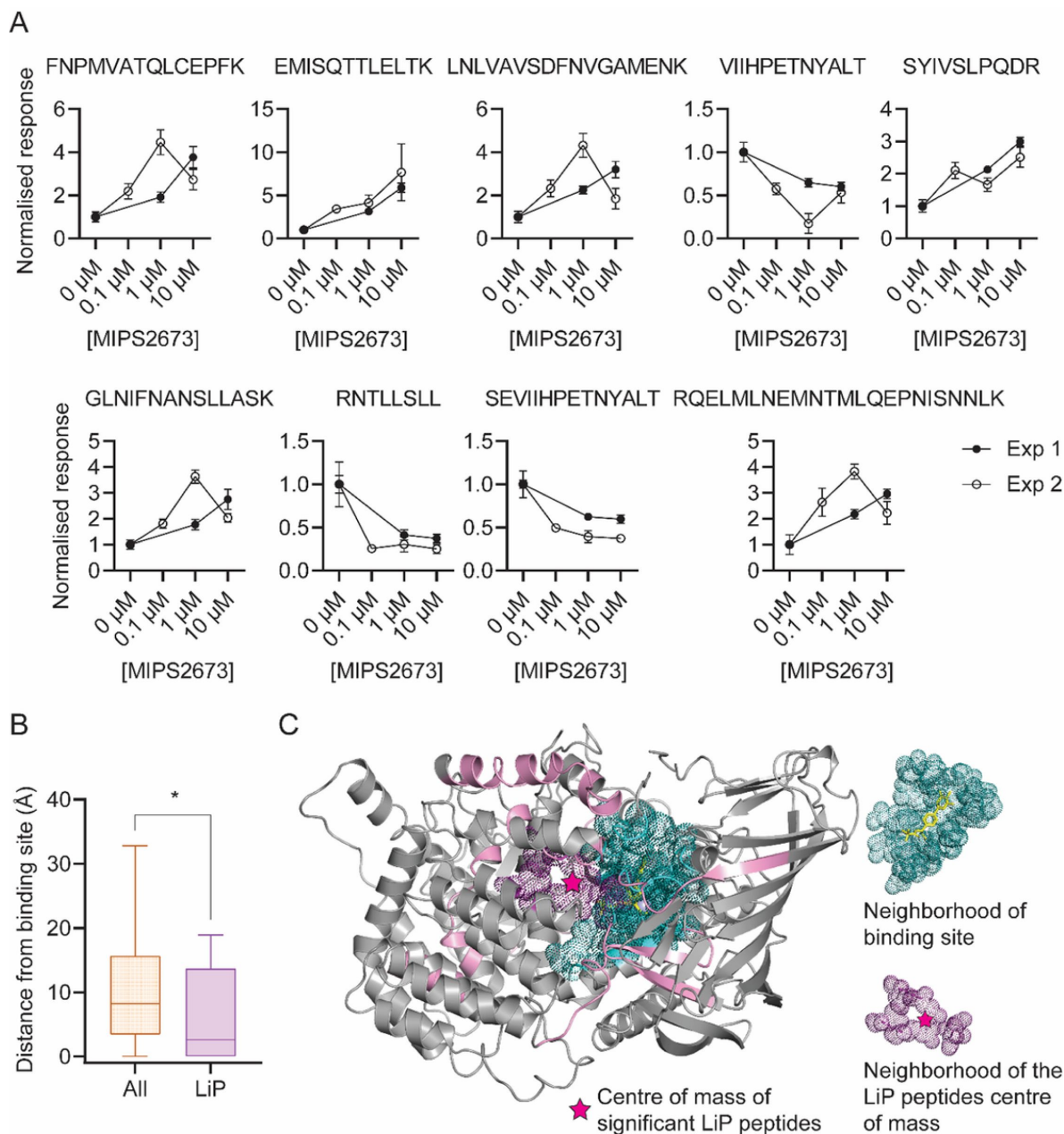


Figure 5

Features of structurally significant PfA-M1 LiP-MS peptides.

(A) Relative abundance of the significant LiP peptides commonly identified across two LiP-MS experiments following *P. falciparum* proteome lysate treatment with MIPS2673. The mean $\pm$ SEM of at least three independent lysate incubations per condition and experiment are shown. (B) Median distances between atoms of the nine significant LiP peptides or all other detected PfA-M1 peptides and the PfA-M1 binding cleft residues. * $p < 0.05$, Mann-Whitney test. (C) MIPS2673 binding site on PfA-M1 determined by X-ray crystallography, and approximation of the MIPS2673 binding site using the significant PfA-M1 LiP peptides and centre of mass calculation. The significant LiP peptides are mapped onto the PfA-M1 structure with MIPS2673 bound and are shown in pink. The drug ligand is shown in yellow and the centre of mass of the significant LiP peptides is shown by a magenta star. The area in cyan represents the neighbourhood of the drug binding site determined as residues within 6.44 Å of bound MIPS2673. The neighbourhood of the LiP peptide centre of mass (residues within 6.44 Å of the centre of mass) is depicted in magenta.

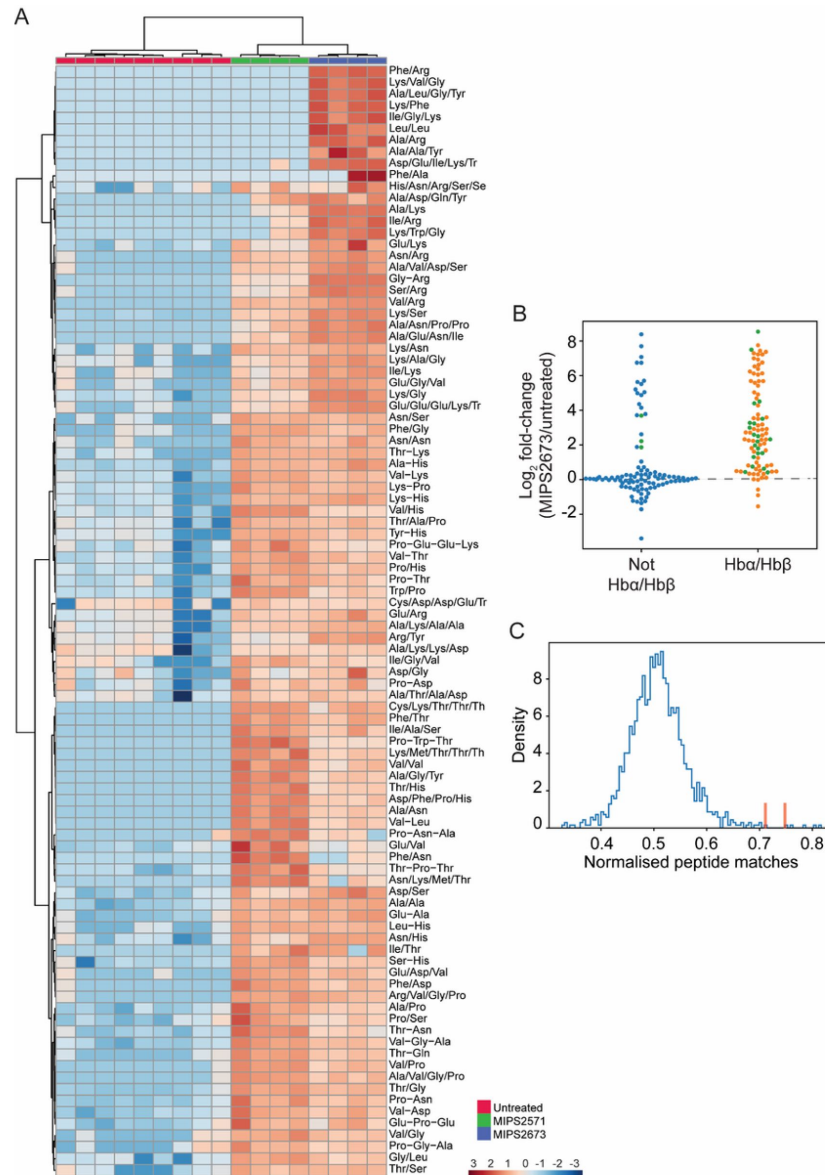


Figure 6

Targeted analysis of significantly dysregulated peptides ($p < 0.05$) following treatment with 1 μM of MIPS2673 (*PfA-M1* inhibitor) for 1 h compared to MIPS2571 (*PfA-M17* inhibitor) and DMSO control.

(A) Hierarchical clustering of the 97 peptides significantly dysregulated after treatment with MIPS2673 (fold-change > 1.5 and $p < 0.05$); 4 biological replicates for MIPS2673 and MIPS2571 (data from Edgar *et al.*, 2022)¹¹ and 9 biological replicates for DMSO control. Vertical clustering displays similarities between samples, while horizontal clusters reveal the relative abundances (median normalized) of the 97 peptides. The color scale bar represents $\log_2$ (mean-centered and divided by the standard deviation of each variable) intensity values. Peptides with hyphen (-) notation indicate confirmed sequence by MS/MS. Peptides with slash (/) notation indicate putative amino acid composition (accurate mass), without confirmed sequence order. (B) Differential enrichment of all (201) putatively identified peptides that could (orange dots) or could not (blue dots) be derived from hemoglobin α , and β . Green dots are peptides that have MS/MS spectra and their sequences have been confirmed. (C) Histogram of the sequence similarity of ~4700 proteins present in *P. falciparum* infected red blood cells to the peptides significantly dysregulated by treatment with MIPS2673. Here, sequence similarity is quantified as the number of times a significantly perturbed peptide matches a given protein, normalized by protein length. The Hb chains α , and β are highlighted in red bars.

potent antimalarials such as artemisinins⁵⁵ and mefloquine⁵⁷. Those antimalarials induce significant depletion of haemoglobin-derived peptides, in contrast to the accumulation observed with the aminopeptidase inhibitors. Overall, the metabolic profile induced by MIPS2673 is unique, and consistent with specific inhibition of PfA-M1.

Discussion

Here, we report the design of MIPS2673, a hydroxamic acid-based inhibitor that targets both the *P. falciparum* (PfA-M1) and *P. vivax* (Pv-M1) M1 aminopeptidases with excellent selectivity over *Plasmodium* M17 and human M1 homologs. We developed orthogonal mass spectrometry-based chemoproteomics methods based on thermal stability and limited proteolysis to validate, in an unbiased manner, that MIPS2673 engages PfA-M1 in a parasite lysate. This was further supported by untargeted metabolomic profiling that revealed significant accumulation of short peptides in MIPS2673-treated parasites, consistent with PfA-M1 inhibition. MIPS2673 is active against a panel of drug resistant *P. falciparum* strains, including parasites with decreased susceptibility to the current frontline antimalarials, has potential to treat both active disease (asexual blood stage activity) and block parasite transmission (anti-gametocyte activity), and is non-cytotoxic to the human HEK293 cell line. Together, our studies confirm *Plasmodium* M1 alanyl aminopeptidase as the target of MIPS2673 and validate selective inhibition of this enzyme as a promising antimalarial strategy.

The gene encoding PfA-M1 was previously shown to be refractory to knockout and is predicted to be essential for blood stage growth, making it an attractive antimalarial drug target.¹² To confirm that MIPS2673 inhibited PfA-M1 in the parasite, we initially developed a streamlined workflow that combined traditional thermal stability proteomics methods with a data-independent acquisition (DIA)-LC-MS/MS approach. Thermal proteome profiling was recently adapted for *P. falciparum* and applied for antimalarial drug target identification using data-dependent acquisition (DDA)-based LC-MS/MS analysis.^{45,53,58} Here, protein thermal stabilisation is typically monitored in the presence or absence of compound over a range of temperatures (thermal proteome profiling), or at multiple drug concentrations at a single temperature (isothermal dose response). The extensive samples generated by these approaches require multiplexing with expensive isobaric stable isotope labelling reagents and offline fractionation to minimise sample complexity prior to DDA-LC-MS/MS analysis. Furthermore, DDA-based proteomics is prone to inconsistent peptide detection, which limits the number of proteins that can be reproducibly identified and quantified. In contrast, using DIA analyses in thermal proteome profiling was recently shown to be an effective and economical alternative to traditional DDA-based thermal shift quantitation workflows.⁵² DIA-based protein quantification also results in high run-to-run reproducibility and a comprehensive dataset that is not biased towards highly abundant proteins.^{43,59} Our DIA-based thermal stability proteomics workflow identified approximately 47% of the detectable *P. falciparum* blood stage proteome,⁵¹ with the undetected proteins primarily comprised of membrane-bound proteins that are not readily extracted with detergent-free buffers required for stability assays of soluble proteins. This proteome coverage is comparable to previously reported DDA thermal proteomics studies of *P. falciparum*,^{45,53,58} but our method avoids the need for testing over a wide range of concentrations or temperatures, isobaric tags and offline fractionation steps that increase the time, cost and complexity associated with previously published methods. Since in bacterial and other eukaryotic cellular systems, limited proteolysis coupled with mass spectrometry (LiP-MS) has proven useful for revealing direct drug targets, protein interactions with endogenous metabolites and ligand binding sites at peptide-level resolution,^{46,54} we also explored LiP-MS based on DIA-LC-MS/MS as a complementary approach to demonstrate MIPS2673 target engagement in malaria parasites.

Both thermal stability proteomics and LiP-MS strategies support unbiased deconvolution of drug mechanisms on a system-wide scale. Whilst our *in vitro* enzyme assay data demonstrated selective inhibition of recombinant *PfA-M1* over *PfA-M17* (4-fold), with a K_i for *PfA-M1* inhibition similar to the EC_{50} for killing *P. falciparum* *in vitro*, this does not rule out inhibition of *PfA-M17* or other protein targets within parasites, especially when the EC_{50} concentration approaches the K_i for *PfA-M17* inhibition.

Thermal stability proteomics and limited proteolysis coupled with global DIA-LC-MS/MS analysis reproducibly identified *PfA-M1* as the sole target of MIPS2673 from approximately 2,000 detected proteins. This validated our chemoproteomics strategy for detection of ligand-protein interactions in complex proteomes and confirmed the on-target activity and M1-selectivity of MIPS2673. Even at 3x EC_{50} , and concentrations expected to inhibit recombinant *PfA-M17*, there were no off-targets consistently identified across both thermal stability proteomics and LiP-MS methods, highlighting the importance of using complementary approaches to map drug interactions and discriminate true protein interactors from false positive identifications.

Our structural analysis showed that MIPS2673 acts by chelating the catalytic zinc ion of the *PfA-M1* active site, similar to all other inhibitors produced to date.^{13,15,19,21,29,33} A strategy targeting the catalytic zinc may present challenges for achieving selectivity over host zinc dependent enzymes, but MIPS2673 demonstrated excellent selectivity over several human M1 homologues, providing rationale for further development of M1 selective inhibitors that present a low risk of host toxicity. Our LiP-MS analysis also supported a zinc-dependent binding mechanism. While other stability-based proteomics methods for drug target identification, such as thermal proteome profiling, map drug-protein interactions by measuring variations in thermal stability of whole proteins, they lack the structural resolution of LiP-MS, which measures local changes in proteolytic susceptibility. Most of the structurally significant *PfA-M1* peptides we identified by LiP-MS were within 5 Å of the MIPS2673 binding site determined by X-ray crystallography. The structurally altered peptides identified with LiP-MS included the peptide $S_{292}EVIHPETNYALT_{305}$, which guards the entrance pathway to the active site and is likely involved in substrate recognition and catalytic competence of the enzyme structure,⁶⁰ as well as the peptide $L_{449}NLVAVSDFNVGAMENK_{465}$, which contains the highly conserved GAMEN exopeptidase motif.³⁰ The GAMEN motif functions to bring substrate to the active site metal for catalysis and contributes hydrogen bonds for ligand binding.³⁰ Interestingly, we also identified several structurally significant LiP-MS peptides distal (> 5 Å) to this location. Ligands and substrates access the *PfA-M1* active site via a long C-terminal channel within domain IV and their migration is orchestrated by several transient states which are stabilized by interactions with specific residues of the domain, including Arg⁹⁶⁹, Arg⁴⁸⁹, Lys⁸⁴⁹, Lys⁹⁰⁷, Glu⁸⁵⁰, Asp⁸³⁰, Glu⁵⁷² and Asp⁵⁸¹.⁶¹ We identified the structurally significant peptides $S_{822}YIVSLPQDR_{831}$ and $R_{898}NTLLSLL_{905}$ that are located > 5 Å from the MIPS2673 binding site at the entrance to the channel cavity. The SYIVSLPQDR and RNTLLSLL peptides contain, or are in very close proximity to, the ASP⁸³⁰ and Lys⁹⁰⁷ residues, respectively, which may have a role in regulating recognition and initial passage and migration of ligands into the channel cavity.⁶¹ Together, this demonstrates the potential of LiP-MS not only for unbiased identification of protein targets of drugs, but also for estimating drug binding and interaction sites.

PfA-M1 has been shown to play an essential role in the terminal stages of haemoglobin digestion and the generation of an amino acid pool within the parasite cytosol.¹⁵ Turnover of haemoglobin into oligopeptides and amino acids peaks during the trophozoite stage when *PfA-M1* expression and activity is highest. In agreement with *PfA-M1* being the target of MIPS2673, we showed that the trophozoite stage coincides with the period that parasites are most susceptible to MIPS2673, and that the growth of MIPS2673-treated parasites becomes stalled at the trophozoite stage. Inhibition of *PfA-M1* using an activity-based probe based on the bestatin scaffold also resulted in trophozoite stage stalling of parasite growth.¹⁵ However, the activity-based probe also caused digestive vacuole swelling, a phenotype that we did not observe. Digestive vacuole

swelling is commonly associated with inhibition of proteases such as the falcipains that are involved in initiating haemoglobin digestion within the parasite digestive vacuole. Since *PfA-M1* is a cytoplasmic protein,²⁸ specific inhibition of this enzyme with MIPS2673 is unlikely to cause digestive vacuole swelling. Instead, the phenotype previously observed with the activity-based probe may be the result of off-target effects. Untargeted metabolomics analysis revealed significant accumulation of short peptides following treatment of parasite cultures with MIPS2673. Many of these accumulated peptides likely originate from haemoglobin as MS/MS and accurate mass isomer analysis shows the sequence of these peptides to be more similar to haemoglobin than other host or parasite proteins. Starving the parasite of key haemoglobin-derived amino acids is therefore the likely mechanism leading to parasite death after MIPS2673 treatment.

Several aminopeptidases in addition to *PfA-M1* are involved in the terminal stages of peptide processing within malaria parasites. These include the essential aminopeptidases, *PfA-M17* and aminopeptidase P (*PfAPP*), and the dispensable enzyme, *PfA-M18*, all of which share a metal-dependent catalytic mechanism. Although no off-targets were consistently identified in our study, four of the seven putative hits identified in our first LiP-MS experiment were metalloaminopeptidases (*PfA-M1*, *PfA-M17*, *PfA-M18* and *PfAPP*), with only *PfA-M1* reproducibly identified as the primary MIPS2673 target across both LiP-MS experiments. *PfA-M18* and *PfAPP* are unlikely to be true off-targets that contribute to the antimalarial mechanism in parasites as we found MIPS2673 is non-inhibitory towards recombinant versions of these enzymes, even at concentrations > 120-fold above the EC₅₀ for *in vitro* parasite killing. We previously reported that specific loss of *PfA-M17* function results in multiple membrane bound digestive vacuoles per parasite. This phenotype was not observed in parasites treated with MIPS2673, arguing against *PfA-M17* being an off-target. Furthermore, distinct peptide profiles for MIPS2673 and a *PfA-M17* inhibitor were observed in our metabolomics dataset, with MIPS2673 showing specific accumulation of basic peptides, which agrees with the broad substrate specificity of *PfA-M1*. In our first LiP-MS experiment we also identified the zinc metalloprotein adenosine deaminase (*PfADA*) as a possible MIPS2673 off-target. *PfADA* is one of three purine salvage enzymes and catalyses deamination of adenosine to inosine, as well as the conversion of 5'-methylthioadenosine, derived from polyamine biosynthesis, into 5'-methylthioinosine. Investigation of metabolites identified in our metabolomics experiment revealed no significant change in purine or polyamine pathway metabolites, suggesting that MIPS2673 has no functional impact on these pathways in parasites. Interestingly, re-analysis of this dataset from experiment one with a less stringent statistical cut-off (Fisher exact test, uncorrected $p < 0.05$) revealed 47 putative targets (Data S4) that were significantly enriched for metallopeptidase ($p = 0.022$), aminopeptidase ($p = 2.1E-4$), aspartic-type endopeptidase ($p = 7.34E-3$) or threonine-type endopeptidase ($p = 1.2E-11$) gene ontology (GO) terms (Fig. S4). Proteins within the metal ion binding gene ontology (GO) term were not significantly enriched ($p = 0.985$) (Data S5), confirming that MIPS2673 does not interact indiscriminately with metalloproteins. Whilst this extended list of putative peptidase-related targets may represent plausible secondary targets – either stabilised by MIPS2673 itself or by peptides accumulating due to primary inhibition of *PfA-M1* – they do not appear responsible for the antiparasitic activity of MIPS2673, as stabilisation of these additional putative targets was not reproducible across independent experiments, and the relevance of the most significant of these (*PfA-M17*, *PfA-M18*, *PfAPP* and *PfADA*) is refuted by our thermal stability, metabolomics and recombinant enzyme inhibition data.

Thermal stability proteomics identified *PfA-M1* and four other proteins that appeared to be stabilised by MIPS2673. Of these four additional proteins, none were metalloprotein or peptidase off-targets, and only the putative AP2 domain transcription factor (PF3D7_1239200) is predicted to be essential for blood stage growth.⁵⁰ This protein is thought to be localised to the parasite nucleus and involved in heterochromatin-associated gene expression.⁶² *PfA-M1* was previously thought to have nuclear localisation⁶³ but is now widely accepted to be cytosolic and no function within the parasite nucleus is currently known. It is unlikely that these proteins contribute to MIPS2673 activity in parasites and it is possible that the cell lysis step required to

generate a parasite lysate for thermal proteomics (and LiP-MS) introduces artefacts by allowing novel protein interactions or compound access to previously compartmentalised proteins. While the proteins identified individually by thermal stability proteomics and LiP-MS could be MIPS2673 off targets, their lack of consistency across assays suggests that they are likely false discoveries. *PfA-M1* was the sole protein reproducibly identified across both unbiased target identification methods, indicating it is the primary target in parasites.

Collectively, we have demonstrated selective targeting of *Plasmodium* M1 alanyl aminopeptidase as an effective antimalarial strategy and validated MIPS2673 as a *PfA-M1* inhibitor through unbiased MS-based chemoproteomics. This work provides a basis for continued development of inhibitors against *Plasmodium* M1 alanyl aminopeptidase in an era when new therapeutics with novel mechanisms of action are urgently needed to combat widespread emergence of malaria parasite drug resistance. More broadly, our study demonstrates the power of mass spectrometry-based techniques to simultaneously probe whole proteomes in an unbiased manner and identify direct drug-protein interactions. The unbiased and orthogonal MS-based proteomics methods described here support investigation of drug mechanisms on a system-wide scale without the need for a specific prior hypothesis about the function of a compound. This makes LiP-MS and thermal stability proteomics ideally suited to reveal the molecular targets of phenotypically active compounds, even for cases where genetic approaches are not tractable. Drug discovery efforts based on phenotypic screens have successfully identified thousands of molecules that kill *Plasmodium* parasites,^{64–66} but most of these hits act by unknown mechanisms. Combined, these complementary target identification strategies will provide greater coverage of the druggable proteome and maximise the chance of deconvoluting targets of compounds with uncharacterised modes of action. These strategies can assist in the discovery of new antimalarial drug targets that will facilitate the development of next-generation medicines with novel and diverse mechanisms, ensuring effective treatments for malaria are available in the future.

Methods

Chemistry

Synthetic Materials and Methods. Chemicals and solvents were purchased from standard suppliers and used without further purification. ^1H NMR, ^{13}C NMR and ^{19}F NMR spectra were recorded on a Bruker Avance Nanobay III 400MHz Ultrashield Plus spectrometer at 400.13 MHz, 100.61 MHz, and 376.50 MHz, respectively. Chemical shifts (δ) are recorded in parts per million (ppm) with reference to the chemical shift of the deuterated solvent. Unless otherwise stated, samples were dissolved in CDCl_3 . Coupling constants (J) and carbon-fluorine coupling constants (J_{CF}) are recorded in Hz and multiplicities are described by singlet (s), doublet (d), triplet (t), quadruplet (q), broad (br), multiplet (m), doublet of doublets (dd), doublet of triplets (dt). LC-MS was performed using either system A or B. System A: an Agilent 6100 Series Single Quad coupled to an Agilent 1200 Series HPLC using a Phenomenex Luna C8 (2) 50 x 4.6 mm, 5 micron column. The following buffers were used; buffer A: 0.1% formic acid in H_2O ; buffer B: 0.1% formic acid in MeCN. Samples were run at a flow rate of 0.5 mL/min for 10 min: 0–4 min 5–100% buffer B in buffer A, 4–7 min 100% buffer B, 7–9 min 100–5% buffer B in buffer A, 9–10 min 5% buffer B in buffer A. Mass spectra were acquired in positive and negative ion mode with a scan range of 100–1000 m/z . UV detection was carried out at 254 nm. System B: an Agilent 6120 Series Single Quad coupled to an Agilent 1260 Series HPLC using a Poroshell 120 EC-C18 50 x 3.0 mm, 2.7 micron column. The following buffers were used; buffer A: 0.1% formic acid in H_2O ; buffer B: 0.1% formic acid in MeCN. Samples were run at a flow rate of 0.5 mL/min for 5 min; 0–1 min 5% buffer B in buffer A, 1–2.5 min 5–100% buffer B in buffer A, 2.5–3.8 min 100% buffer B, 3.8–4 min 100–5% buffer B in buffer A, 4–5 min 5% buffer B in buffer A. Mass spectra were acquired in positive and negative ion mode with a scan range of 100–1000 m/z . UV detection was carried out at 214 and 254 nm.

Methyl 2-(2-hydroxy-2-methylpropanamido)-2-(3',4',5'-trifluoro-[1,1'-biphenyl]-4-yl)acetate (3). A mixture of methyl 2-amino-2-(4-bromophenyl)acetate (2) (400 mg, 1.64 mmol), 2-hydroxy-2-methylpropanoic acid (205 mg, 1.97 mmol), EDCI (1.2 eq) and DMAP (1.3 eq) in DCM (10 mL/mmol of amine) was stirred at room temperature under nitrogen overnight. The mixture was diluted with DCM, washed with 2 M HCl_(aq), saturated NaHCO_{3(aq)}, then brine. The organic layer was dried over Na₂SO₄, filtered and concentrated under reduced pressure. The product was subsequently reacted with 3,4,5-trifluorophenylboronic acid (347 mg, 1.97 mmol), Pd(PPh₃)₂Cl₂ (35 mg, 0.05 mmol), Na₂CO₃ (1 M, 3.3 mL) in THF (9.9 mL) in a sealed microwave vial and heated at 100 °C for 2 h. After cooling, the mixture was diluted with EtOAc (10 mL) and water (10 mL), and the mixture extracted with EtOAc. The organic layer was dried over Na₂SO₄, filtered and concentrated. The crude product was purified by column chromatography to give compound 3 (503 mg, 80%) as a colourless oil. ¹H NMR δ 7.73 (br. d, *J* = 7.0 Hz, 1H), 7.53–7.39 (m, 4H), 7.22–7.06 (m, 2H), 5.57 (d, *J* = 7.3 Hz, 1H), 3.76 (s, 3H), 1.50 (s, 3H), 1.44 (s, 3H); ¹⁹F NMR δ -133.85 (d, *J* = 20.5 Hz), -162.09 (dd, *J* = 20.5 Hz); ¹³C NMR δ 175.9, 171.1, 151.6 (ddd, *J*_{CF} = 250.3/10.2/4.6 Hz), 141.2–138.0 (m, 2C), 136.9, 136.8–136.5 (m), 128.1, 127.7, 111.6–110.9 (m), 73.8, 56.1, 53.1, 28.0, 27.9; LC-MS *t*_R: 3.4 min, *m/z* 381.9 [MH]⁺.

2-Hydroxy-N-(2-(hydroxyamino)-2-oxo-1-(3',4',5'-trifluoro-[1,1'-biphenyl]-4-yl)ethyl)-2-methylpropanamide [MIPS2673 (4)]. Methyl 2-(2-hydroxy-2-methylpropanamido)-2-(3',4',5'-trifluoro-[1,1'-biphenyl]-4-yl)acetate (3) (100 mg, 0.26 mmol) was dissolved in anhydrous MeOH (5 mL/mmol of ester) at room temperature. NH₂OH.HCl (4.0 eq) was added followed by KOH (5M in anhydrous MeOH, 5.0 eq). The mixture was stirred at room temperature overnight and monitored by LC-MS analysis. The mixtures were directly dry-loaded on to Isolute HM-N[®] (Biotage), before purification by FCC. The desired hydroxamic acid 3 was obtained as white solid (40 mg, 40% yield). ¹H NMR (DMSO-*d*₆) δ 11.13 (s, 1H), 9.15 (s, 1H), 8.07 (d, *J* = 8.1 Hz, 1H), 7.78–7.63 (m, 4H), 7.46 (d, *J* = 8.3 Hz, 2H), 5.74 (s, 1H), 5.29 (d, *J* = 8.1 Hz, 1H), 1.26 (s, 3H), 1.23 (s, 3H); ¹⁹F NMR (DMSO-*d*₆) δ -134.91 (d, *J* = 21.7 Hz), -163.45 (dd, *J* = 21.8 Hz). ¹³C NMR (DMSO-*d*₆) δ 176.2, 166.4, 150.9 (ddd, *J*_{CF} = 246.8/9.8/4.3 Hz), 139.5, 140.2–137.1 (m), 136.9–136.4 (m, 2C), 127.3, 127.2, 111.8–110.9 (m), 72.5, 53.2, 27.8, 27.6; *m/z* HRMS (TOF ES⁺) C₁₈H₁₈F₃N₂O₄ [MH]⁺ calcd 383.1213; found 383.1214; LC-MS *t*_R: 3.2 min; HPLC *t*_R: 7.7 min, > 99%.

Routine parasite culture

P. falciparum parasites (3D7 line) were cultured continuously in O + human red blood cells (Australian Red Cross Blood Service) at 2–4% haematocrit using standard methods,⁵⁷ with minor modifications.⁵⁵ Briefly, cultures were maintained at 37°C under defined atmospheric conditions (94% N₂, 5% CO₂, and 1% O₂) in RPMI 1640 medium supplemented with 0.5% Albumax II (Gibco, Australia), HEPES (5.94 g/L), hypoxanthine (50 mg/L) and sodium bicarbonate (2.1 g/L). Parasites were synchronised using multiple rounds of sorbitol lysis⁶⁸ and monitored at least every 48 h by Giemsa staining of methanol-fixed blood smears.

Parasite material for thermal stability and limited proteolysis experiments was obtained from cultures synchronised to 30–38 h post invasion. Parasites were isolated from red blood cells via hypotonic lysis to avoid the use of solubilising molecules such as saponin. Briefly, infected red blood cells were magnetically purified and incubated with 10 mL of hypotonic lysis buffer (150 mM ammonium chloride, 10 mM potassium bicarbonate, 1 mM EDTA in Milli-Q water) for 5 min at 4°C. Following hypotonic lysis, the intact parasite pellets were washed three times in PBS and stored at -80°C until whole proteome preparation.

Whole proteome preparation

Parasite pellets were resuspended in 1–2 mL of cold 100 mM HEPES buffer (pH 8.1) and subjected to freeze-thaw lysis by cycling at 2–3 min intervals between dry ice and a 35°C heat block. Cells used for limited proteolysis also underwent mechanical shearing after freeze-thaw lysis by passing the thawed suspension through a 26-gauge (10-20x) and then a 30.5-gauge needle (10-20x). The

resulting lysates were centrifuged to remove cell debris (20,000 *g* for 20 min at 4°C), the supernatants collected and combined to create one pooled parasite lysate. For thermal stability experiments, the cell debris obtained after centrifugation underwent two additional cycles of freeze-thaw and the collected supernatants were added to the pooled lysate. All remaining steps were performed at 4°C. The protein concentration of the resulting lysate was determined using the Pierce bicinchoninic acid (BCA) protein assay kit (ThermoFisher Scientific).

Lysate treatment for thermal stability proteomics and MS analysis

The pooled lysate was equally separated to achieve at least three incubations (technical replicates) per condition. The parasite lysates were incubated with 1 μ M or 4 μ M of MIPS2673 or vehicle (DMSO) for 3 min at room temperature, followed by thermal challenge at 60°C for 5 min. Each sample was incubated at room temperature for a further 3 min prior to being returned to ice. The denatured protein was removed via ultracentrifugation at 100,000 *g* for 20 min (4°C) in a Beckman Coulter Optima XE-90 – IVD ultracentrifuge with a 42.2 Ti rotor. The protein concentration in the soluble fraction was determined using a BCA assay. Samples were then reduced and alkylated at 95°C for 5 min using 10 mM tris(2-carboxyethyl)phosphine hydrochloride (TCEP) and 40 mM iodoacetamide (40 mM final concentration), respectively. Each sample was digested overnight with MS-grade trypsin (1:50; Promega) at 37°C while shaking. The following day, trypsin activity was quenched using 5% formic acid (FA) and samples were subjected to desalting using in-house-generated StageTips, as described previously.⁶⁹ The samples were then dried and resuspended in 12 μ L of 2% (v/v) acetonitrile (ACN) and 0.1% (v/v) FA containing indexed retention time (iRT) peptides (Biognosys) for LC-MS/MS analysis.

Lysate treatment for limited proteolysis and MS analysis

An equivalent volume of protein lysate was aliquoted from a lysate pool for a minimum of three independent incubations per condition and incubated with MIPS2673 (0.04 μ M to 40 μ M) or vehicle (DMSO) at room temperature for 10 min. Proteinase K from *Tritirachium album* (Sigma Aldrich) was added to all of the reactions at a 1:100 ratio of enzyme to protein and incubated at room temperature for 4 min. The digestion reactions were stopped by heating samples to 98°C for 1 min, followed by addition of an equal volume of 10% sodium deoxycholate (Sigma Aldrich) to achieve a final concentration of 5%. Samples were incubated for a further 15 min at 98°C. These samples were then removed from the heat and subjected to complete digestion under denaturing conditions.

Samples were reduced and alkylated for 10 min at 95°C with 10 mM TCEP and 40 mM chloroacetamide, respectively. Subsequently, samples were digested with Trypsin/LysC (1:100 enzyme to substrate ratio) for 16 h at 37°C while shaking. Deoxycholate was precipitated by addition of formic acid to a final concentration of 1.5% and centrifuged at 16,000 *g* for 10 min. After transferring the supernatant to a new tube an equal volume of FA was added again and the centrifugation repeated. Digests were desalted using in-house-generated StageTips⁶⁹ and after drying resuspended in 12 μ L of 2% (v/v) ACN and 0.1% (v/v) FA containing iRT peptides (Biognosys) for LC-MS/MS analysis.

LC-MS/MS acquisition for proteomics

LC-MS/MS was carried out using data-independent acquisition mode on a Q-Exactive HF mass spectrometer (Thermo Scientific) as described previously.⁵¹ Briefly, samples were loaded at a flow rate of 15 μ L/min onto a reversed-phase trap column (100 μ m x 2 cm), Acclaim PepMap media (Dionex), maintained at a temperature of 40°C. Peptides were eluted from the trap column at a flow rate of 0.25 μ L/min through a reversed-phase capillary column (75 μ m x 50 cm) (LC Packings, Dionex). The HPLC gradient was 158 min and gradually reached 30% ACN after 123 min, 34% ACN after 126 min, 79% ACN after 131 min and 2% after 138 min for a further 20 min. The mass spectrometer was operated in a data-independent mode with a 43-fixed-window setup of 18 *m/z*

effective precursor isolation over the m/z range of 364–988 Da. Full scan was performed at 60,000 resolution (AGC target of $3e^6$ and maximum injection time of 54 ms) with fragmentation resolution at 15,000 (AGC target of $2e^5$, maximum injection time of 22 ms, normalized collision energy of 27.0).

Data analysis for thermal stability proteomics experiments

Raw files were processed using Spectronaut™ 13.0 against an in-house generated *P. falciparum* infected red blood cell spectral library as described previously.⁵¹ The consensus library contained 42,245 peptides corresponding to 4,421 protein groups. For processing, raw files were loaded and Spectronaut calculated the ideal mass tolerances for data extraction and scoring based on its extensive mass calibration with a correction factor of 1. Both at precursor and fragment level, the highest data-point within the selected m/z tolerance was chosen. Identification of peptides against the library was based on default Spectronaut settings (Manual for Spectronaut 13.0, available on Biognosis website). Briefly, precursor Qvalue Cut-off and Protein Qvalue Cut-off were as per default at 1%, therefore only those that passed the cut-off were considered as identified and used for subsequent processing. Retention time prediction type was set to dynamic iRT. Interference correction was on MS2 level. For quantification, the interference correction was activated and a cross run normalisation was performed using the total peak area as the normalisation base with a significance level of 0.01.

Significantly stabilised proteins were determined by first calculating the fold-change between the MIPS2673 (1 μ M and 4 μ M conditions) and vehicle-treated samples for each experiment. Statistical significance of the change in protein abundance was determined by a Welch's t-test. Proteins that were significantly stabilised (fold-change > 1.2, $p < 0.05$) at both MIPS2673 concentrations in multiple experiments were considered putative targets. The data were plotted using paired volcano plots, whereby fold-change in protein abundance was plotted as a function of statistical significance for multiple conditions or experiments to visualise commonly stabilised proteins.

Data analysis for limited proteolysis experiments

For limited proteolysis experiments, raw data files were processed using Spectronaut™ 13.0 as described above, except a limited proteolysis-specific *P. falciparum* infected red blood cell spectral library was used. The in-house generated library contained 85,039 peptides corresponding to 4,681 protein groups.

All limited proteolysis datasets were analysed at the modified peptide sequence level. The datasets were first analysed for differentially abundant peptides between MIPS2673 and vehicle by filtering based upon relative peptide abundance (absolute fold-change > 1.5) and statistical significance ($q < 0.01$). Within independent experiments, peptides meeting these thresholds at each drug concentration were considered significant. To filter and prioritise possible protein targets, a one-sided Fisher exact test was used to determine whether, based on the total number of peptides detected for each protein in the dataset, the number of significant LiP peptides identified for each protein was greater than would be expected by chance. Resulting p-values were adjusted for multiple testing using the Bonferroni method. Proteins passing the Bonferroni corrected statistical cut-off were considered putative targets. Proteins that did not pass the stringent Bonferroni corrected statistical cut-off, but were statistically significant ($p < 0.05$) were considered lower confidence targets. The position of drug binding using the centre of mass and distance measurements were performed using PyMol 2.5.1 (Schrodinger).

To reveal in an unbiased way whether any protein functions are overrepresented in the expanded list of candidate targets from the experiment one dataset, this list was tested for functional enrichments using the *topGO* R-package.⁷⁰ All proteins with an uncorrected $p < 0.05$ were tested against a background of proteins for which abundance was measured. PlasmoDB GO terms were used for GO term mapping. We tested for enrichment in molecular function GO terms with the

Fisher exact test using the weight-algorithm in topGO.⁷¹ All terms with a p-value less than 0.05 were considered significant. GO terms represented by fewer than five proteins in the background dataset were excluded from the analysis.

Enzyme assays

Inhibition of aminopeptidase activity assays were performed against nine aminopeptidases: the M1 alanyl and M17 leucyl aminopeptidases from *Plasmodium falciparum* (PfA-M1; PfA-M17) and *Plasmodium vivax* (Pv-M1; Pv-M17), the M18 aspartyl and aminopeptidase P aminopeptidases from *Plasmodium falciparum* (PfA-M18; PfAPP) and three human M1 homologues: LTA4H (OriGene TP307617), ERAP1 (OriGene TP314469) and ERAP2 (Creative BioMart ERAP2-304H). The *Plasmodium* enzymes were produced recombinantly as described previously.^{30,34,72–74} and the human recombinant enzymes were purchased from commercial suppliers as indicated.

The ability of MIPS2673 to inhibit aminopeptidase activity was assessed by fluorescence assays using fluorogenic substrates *L*-Leucine-7-amido-4-methylcoumarin hydrochloride (H-Leu-NHMec) for all enzymes except LTA4H that was assessed using *L*-Alanine-7-amido-4-methylcoumarin hydrochloride (H-Ala-NHMec, 20 μ M) and PfAPP that was assessed using Lys(Abz)-Pro-Pro-NA (PPpNA, 10 μ M).²⁰ Aminopeptidase activity of Pf-M18 was measured by an absorbance assay using the substrate *L*-glutamic acid γ -(4-nitroanilide) (L-Glu-pNA, 100 μ M).⁷⁴ The concentration of substrate in the assay was held constant for each enzyme and ranged from 10–100 μ M depending on the enzyme. Reactions were measured at 37°C in white 384-well plates in a total volume of 50 μ L using a spectrofluorimeter (BMG FLUOstar) with excitation at 355 nm and emission at 460 nm for the fluorescence assays or excitation at 405 nm for the absorbance assays. The fluorescence or absorbance signal was continuously monitored until a final steady state velocity, v , was obtained. Inhibition constants were calculated in biological triplicate from three different protein preparations. For determination of the apparent Morrison inhibition constant (denoted here as K_i), enzymes were pre-incubated in 100 mM Tris-HCl, pH 8.0 (supplemented with 1 mM CoCl₂ for M17 aminopeptidases) and inhibitor added 20 min prior to the addition of substrate. Substrate concentrations were selected to allow sensitive detection of enzyme activity while not exceeding the K_m for each enzyme. A compound concentration range was selected to obtain a complete inhibition curve (0% – 100%) in biological triplicate. Where possible, the K_i values were calculated by plotting the initial rates versus inhibitor concentration, and fitting to the Morrison equation for tight-binding inhibitors in GraphPad Prism (non-linear regression method). Where a K_i could not be calculated (ERAP1), the percentage inhibition was calculated assuming 100% activity in the absence of compound.

Structural biology

PfA-M1 was co-crystallized with MIPS2673 by the hanging-drop method using previously established protocols.³⁰ Purified protein was concentrated to 5.0 mg/mL and mixed with MIPS2673 to a final ligand concentration of 1 mM. Crystals grew in 20 – 30% poly(ethylene glycol) (PEG)8000, 0.1 M Tris pH 7.5 – 8.5, 0.2 M MgCl₂, 10% glycerol. Co-crystals were subjected to an additional overnight compound soak before being snap-frozen in liquid nitrogen. Data were collected at 100 K using synchrotron radiation at the Australian Synchrotron beamlines 3BM1.⁷⁵ Data were processed using XDS,⁷⁶ and Aimless,⁷⁷ as part of the CCP4i program suite.⁷⁸ The structures were solved by molecular replacement in Phaser⁷⁹ using the structure of unliganded PfA-M1 coordinates (RCSB ID 3EBG) as the search models. The structures were refined using Phenix,⁸⁰ with 5% of reflections set aside for calculation of R_{free} . Between refinement cycles, the protein structure, solvent, and inhibitors were manually built into $2F_o - F_c$ and $F_o - F_c$ electron density maps using COOT,^{81,82} with restraint files generated by Phenix where necessary. The data collection and refinement statistics can be found in Supplemental Information (Table S5). The coordinates and structure factors are available from the Protein Data Bank with PDB Accession codes: 8SLO.pdb (PfA-M1-MIPS2673).

Determination of compound EC₅₀ against 3D7 *P. falciparum* parasites

Parasite viability assay was performed as previously described.¹¹ Synchronized ring stage *P. falciparum* 3D7 parasites were cultured in 96-well U-bottom plates at 0.5% parasitemia and 2% haematocrit, to which 50 µL of serially diluted MIPS2673 or artesunate was added. After 72 h plates were placed at -80°C before being analysed in buffer containing 20 mM Tris pH 7.5, 5 mM EDTA, 0.008% saponin (w/v) and 0.008% Triton x-100 (v/v)⁸³ containing 0.2 µL/mL SYBR Green I Nucleic Acid Gel Stain (10,000x in DMSO; ThermoFisher). After 1 h incubation, fluorescence intensity was read on a Glomax® Explorer Fully Loaded plate reader (Promega) at emission wavelengths of 500–550 nm and an excitation wavelength of 475 nm. Uninfected RBCs and parasites treated with DMSO were used to normalize fluorescence. Data from 4 biological replicates performed in triplicate were plotted as 4-parameter log dose nonlinear regression analysis with a sigmoidal dose-response curve fitted using GraphPad Prism 9 to generate EC₅₀ values, with error bars representative of the SEM.

IC₅₀ determination against drug resistant field isolates and laboratory selected *P. falciparum* parasites

In vitro testing was performed with a modified [³H]-hypoxanthine incorporation assay, as previously reported.⁸⁴

Parasite killing rate assay

Assay was performed as previously described.¹¹ Synchronized ring or trophozoite *P. falciparum* 3D7 parasites were cultured in the presence of 10x the EC₅₀ of MIPS2673 or artesunate for either 24 or 48 h. Media containing compound was replenished at 24 h for 48 h assays. After incubation, RBCs were thoroughly washed to remove the compound and cultures were diluted 1/3 with fresh media and further grown for 48 h before being placed at -80 °C. Cultures were thawed and analyzed using SYBR Green I assay as described above. Parasite viability was determined as a percentage of DMSO-treated parasites cultured alongside compound treated parasites. Artesunate was used as a comparison of the parasite killing rate, and experiments were performed in 4 biological replicates

Gametocyte activity assay

Viability assays were performed as previously described^{85–87} against *PfNF54-s16-GFP* early (I-III); late (IV-V) and mature (V) stage gametocytes. Gametocytes were assessed at early stage (day 2), late stage (day 8) and mature (day 12). Compounds diluted in 4% DMSO were transferred into 384-well imaging plates; gametocytes at various stages added, and plates incubated for 72 h in 5% CO₂, 5% O₂ and 60% humidity at 37°C. After 72 h incubation, 5 µL of MitoTracker Red CMH2XROS in phosphate buffered saline (PBS) added per well and plates incubated overnight. Image acquisition and analysis was undertaken on the Opera QEHS micro-plate confocal imaging system. An Acapella based script using the MTR fluorescent signal and the GFP designated object quantifying viable stage dependent parasite morphology identified gametocytes. Gametocyte viability was calculated as a percentage of the positive (5 µM puromycin) and negative (0.4% DMSO) controls. EC₅₀ values were calculated using a 4-parameter log dose, non-linear regression analysis, with sigmoidal dose response (variable slope) curve fit using Graph Pad Prism (ver 4.0). No constraints were used in the curve fit. Chloroquine, artesunate, pyronaridine, pyrimethamine, dihydroartemisinin and methylene blue were used as control compounds. Experiments were completed in duplicate for 2 or 3 biological replicates.

HEK293 cytotoxicity assay

To assess the cytotoxicity of MIPS2673 against mammalian cells, an established and well validated metabolic assay for HEK293 cells was used.⁸⁸ In brief, HEK293 cells were maintained in DMEM medium supplemented with 10% FBS. Compound testing was undertaken in tissue culture-treated 384-well plates incubated for 72 h at 37 °C and 5% CO₂. The media was then removed and replaced with resazurin (final concentration 44 µM). After an additional 6 h incubation, total fluorescence was measured using an Envision plate reader (Perkin Elmer) (excitation/emission: 530 nm/595 nm).

Metabolomics

P. falciparum 3D7 cultures that underwent double sorbitol synchronisation were incubated for a further 28–42 h to achieve the desired trophozoite stage (28 hours post invasion (h.p.i)) at 6% parasitaemia and 2% haematocrit. Infected RBCs (2×10^8) were treated with 3 x the EC₅₀ of MIPS2673 (1 µM) or an equivalent volume of DMSO (negative control) for 1 h, after which metabolites were extracted. Following incubation, all samples were centrifuged at 650 g for 3 min, supernatants were removed, and pellets were washed in 1 mL of ice-cold PBS. Samples were again centrifuged at 650 g for 3 min to remove all of the PBS and the pellets were resuspended in 150 µL of ice-cold extraction buffer (100% methanol). Samples were then incubated on an automatic vortex mixer for 1 h at 4°C before being centrifuged at 21,000 g for 10 min. Supernatant was collected and 100 µL was stored at – 80°C until further analysis and 10 µL from each sample was pooled to serve as a quality control (QC) sample. Liquid chromatography coupled to mass spectrometry (LC-MS) analysis and data processing were as previously described.¹¹

Kinetic solubility estimation

Kinetic solubility was estimated using nephelometry (Sol_{pH}) according to the method of Bevan and Lloyd.⁸⁹ Briefly, the compound in DMSO was spiked into either pH 6.5 phosphate buffer or 0.01 M HCl (approximately pH 2) with the final DMSO concentration being 1%. After 30 min had elapsed, samples were then analysed via nephelometry to determine a solubility range.

Declarations

Significance

Drug resistance now threatens all existing treatments for malaria, the world's most lethal parasitic disease of humans. As a result, there is an urgent need for new antimalarial drug candidates that act via novel pathways and have a high barrier for resistance. It is therefore important to understand the mechanism of action for emerging antimalarials. However, target deconvolution is a major bottleneck for new drug development for malaria, particularly when genetic approaches for target identification are not possible. Mass spectrometry-based chemoproteomics has emerged as a powerful alternative strategy to deconvolute drug protein targets in a proteome-wide and unbiased manner. To this end, we report a novel promising antimalarial compound targeting M1 alanyl aminopeptidase (PfA-M1) in the malaria parasite, and validate this target by combining biochemical assays on recombinant enzymes, with metabolomics, thermal stability chemoproteomics and a novel drug target deconvolution strategy for *Plasmodium falciparum*: limited proteolysis coupled mass spectrometry (LiP-MS). LiP-MS uses mass spectrometry to detect proteolytic stabilisation of proteins in a parasite lysate due to ligand binding. LiP-MS also estimated the binding site of MIPS2673 on PfA-M1 to within ~5 Å of that determined by X-ray crystallography, highlighting the value of this approach in resolving both the molecular target and binding site of drugs. Implementing LiP-MS in *P. falciparum* expands the drug target deconvolution toolbox for malaria drug discovery. Our studies chemically validated selective targeting of *Plasmodium* M1 alanyl aminopeptidase as a promising multi-stage and cross-species strategy for the treatment of malaria and demonstrated the value of combining multiple orthogonal

mechanism of action studies for target deconvolution. This combination of methods provides a validated workflow to help uncover the targets of other promising antimalarial candidates in the pipeline.

Author contributions

Conceptualization, C.G., M.P.C., T.F.dK-W., P.J.S., S.M., and D.J.C.; Formal Analysis, C.G., M.P.C., G.S., R.C.S.E., T.R.M., C.T.W., N.D., C.A.M., S.D., S.W., and S.M.; Investigation, C.G., M.P.C., G.S., R.C.S.E., T.R.M., C.T.W., N.D., N.B.V., N.A.C., and S.D.; Resources, S.W., V.M.A., T.F.dK-W., P.J.S., S.M., and D.J.C.; Data Curation, C.G., M.P.C., G.S., R.C.S.E., T.R.M., C.T.W., N.D., N.B.V., S.D., and S.W.; Writing – Original Draft, C.G., G.S., P.J.S., S.M., and D.J.C.; Writing – Review & Editing, C.G., M.P.C., G.S., R.C.S.E., C.T.W., N.B.V., C.A.M., S.W., S.M.D., V.M.A., T.F.dK-W., P.J.S., S.M., and D.J.C.; Visualisation, C.G., M.P.C., G.S., R.C.S.E., C.A.M., P.J.S., and S.M.; Supervision, S.M.D., V.M.A., T.F.dK-W., P.J.S., S.M., and D.J.C.; Project Administration, C.G., T.F.dK-W., P.J.S., S.M., and D.J.C.; Funding Acquisition, T.F.dK-W., P.J.S., S.M., and D.J.C.

Data and code availability

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE⁹⁰ [partner repository \(https://www.ebi.ac.uk/pride/archive/\)](https://www.ebi.ac.uk/pride/archive/) with the dataset identifier PXD044125.

Reviewer account details:

Username: reviewer_pxd044125@ebi.ac.uk

Password: XgW6Kph8

The mass spectrometry metabolomics data is available at the NIH Common Fund's National Metabolomics Data Repository (NMDR) website, the Metabolomics Workbench, <https://www.metabolomicsworkbench.orgwhere> it has been assigned Project ID (ST002792). The data can be accessed directly via its Project DOI: 10.21228/M8SQ7X. This work is supported by Metabolomics Workbench/National Metabolomics Data Repository (NMDR) (grant# U2C-DK119886), Common Fund Data Ecosystem (CFDE) (grant# 3OT2OD030544) and Metabolomics Consortium Coordinating Center (M3C) (grant# 1U2C-DK119889)

Structural data has been deposited with PDB with the identifier 8SLO.

This paper does not report original code.

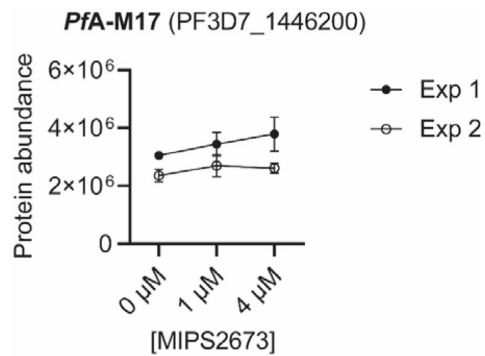
Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Acknowledgements

The authors wish to thank and acknowledge the Australian Red Cross LifeBlood for the provision of fresh red blood cells, without which antimalarial testing could not have been performed. The Centre for Drug Candidate Optimisation (CDCO, Monash University) is acknowledged for providing the solubility data. The CDCO is partially supported by the Monash University Technology Research Platform network and Therapeutic Innovation Australia (TIA) through the Australian Government National Collaborative Research Infrastructure Strategy (NCRIS) program. We thank the following colleague for support with the *Plasmodium falciparum* strain panel assays: Sibylle Sax, Swiss Tropical and Public Health Institute, Switzerland. We thank the Monash Proteomics and Metabolomics Facility for technical support. S.D. was supported by a NHMRC Dora Lush Biomedical

Postgraduate Scholarship (AP1150359), Griffith University DVCR Postgraduate Top-up Scholarship and a Discovery Biology Top-up Scholarship. V.M.A acknowledges Medicines for Malaria Venture for continued financial support. Funding was provided by an NHMRC Synergy Grant for the Antimalarial Synergy Team #1185354.

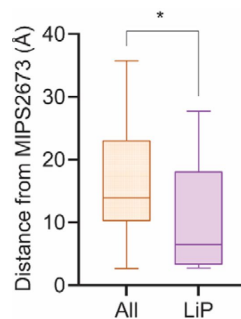
Supplementary Materials



Supplemental Fig. S1

Individual thermal stabilisation profile of *PfA*-M17 leucyl aminopeptidase after *P. falciparum* lysate treatment with MIPS2673. Related to Fig. 3 [↗](#).

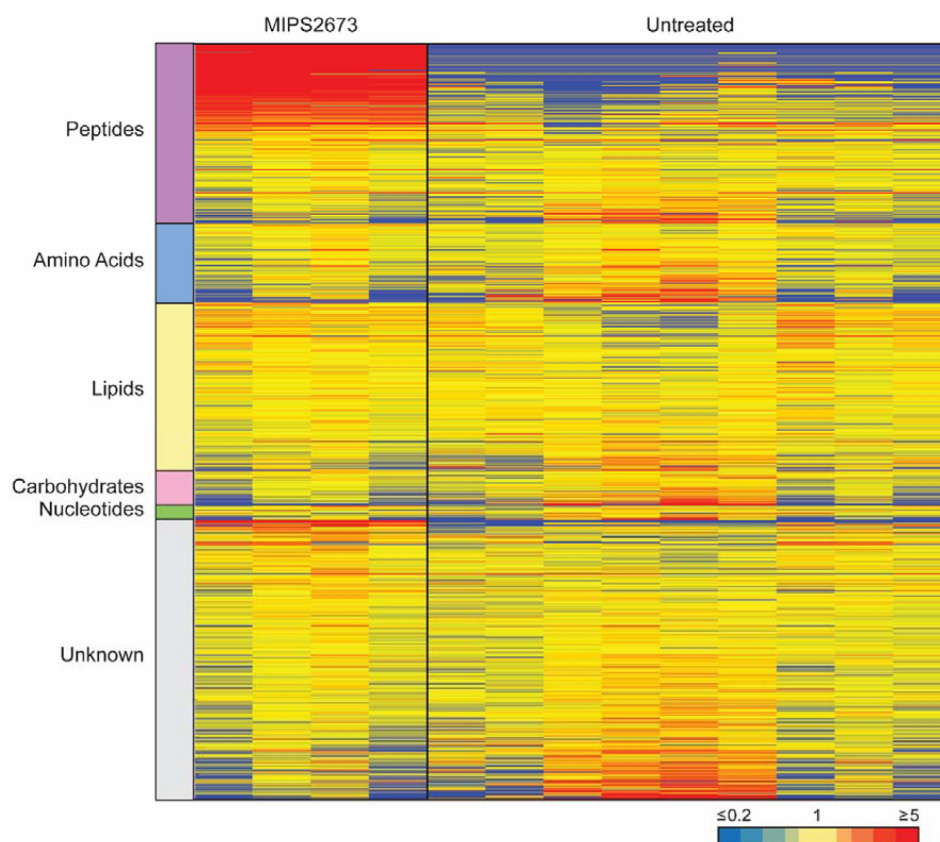
Abundance of *PfA*-M17 leucyl aminopeptidase from *P. falciparum* lysates in the absence and presence of MIPS2673 after a 60 °C thermal challenge. *PfA*-M17 was not stabilised by MIPS2673 after a 60 °C thermal challenge ($p > 0.05$). Error bars indicate SD ($n = 4$ independent lysate incubations per condition in each independent experiment).



Supplemental Fig. S2

Minimum distance of significant LiP-MS or all other *PfA*-M1 peptides from bound MIPS2673. Related to Fig. 5 [↗](#).

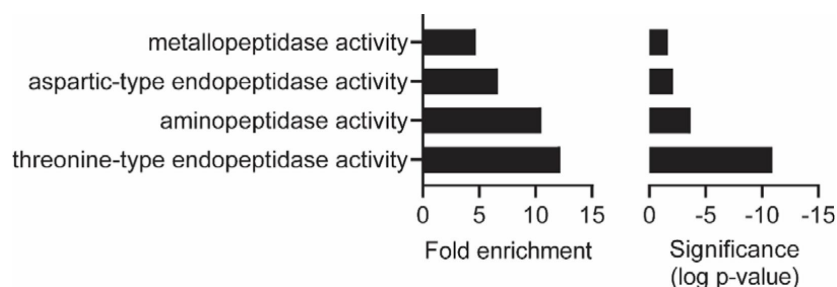
Among the 108 *PfA*-M1 peptides commonly identified across both LiP-MS experiments, nine peptides were significantly dysregulated in the presence of MIPS2673 at all concentrations tested. The atoms of these LiP peptides were used to measure the distribution of distances to bound MIPS2673 compared to all other detected *PfA*-M1 peptides. * $p < 0.05$.



Supplemental Fig. S3

Untargeted metabolomics analysis of 3D7 parasites treated with 1 μ M of MIPS2673. Related to Fig. 6.

Heatmap analysis of peak intensities of all putative metabolites for each condition; 1 μ M of MIPS2673 for 1 h and DMSO control (untreated). Data is shown from 4-9 biological replicates, red, blue and yellow indicates increase, decrease and no change, respectively in the relative abundance of putative metabolites identified.



Supplemental Fig. S4

Molecular function GO enrichment analysis of the MIPS2673 experiment one LiP-MS dataset. Related to Fig. 4.

Fold enrichment (left) and significance (right) of GO molecular functions enriched among the expanded list of candidate MIPS2673 targets identified in the first LiP-MS experiment. All proteins with an uncorrected $p < 0.05$ were tested against a background of proteins for which abundance was measured. PlasmoDB GO terms were used for GO term mapping. Enrichment in molecular function GO terms was tested with the Fisher exact test using the weight-algorithm in topGO.⁷¹ All terms with a p -value less than 0.05 were considered significant. GO terms represented by fewer than five proteins in the background dataset were excluded from the analysis.

Concentration (μ M)	% inhibition (%) ^a				
	<i>Pf</i> -M1	<i>Pv</i> -M1	LTA4H	ERAP1	ERAP2
1.25	100	100	0	0	31
10	100	100	58	2.3	0
500	-	-	42.2	89.8	97.8
1000	-	-	53.4	95.6	98.7

^a Values represent the mean of three independent experiments. LTA4H, leukotriene A-4 hydrolase. ERAP1, endoplasmic reticulum aminopeptidase 1. ERAP2, endoplasmic reticulum aminopeptidase 2

Supplemental Table S1

Related to Fig. 1 [↗](#).

Percent inhibition by MIPS2673 of *Pf*A-M1 and *Pv*-M1 aminopeptidases compared to selected human M1 homologues.

Compound	% inhibition ^a	
	40 μ M	120 μ M
MIPS2673	NI	89
Puromycin^b	100	103
Chloroquine	67	65
DHA	29	32

^a Values represent the mean from 2 experiments. ^b Puromycin was tested at 10 μ M. NI, no inhibition.

Supplemental Table S2

Related to Fig. 1 [↗](#).

Cytotoxicity of MIPS2673 against the human HEK293 cell line.

<i>P. falciparum</i> strain	Sensitivity profile	IC ₅₀ (nM) ^b	Fold shift in IC ₅₀ relative to	
			NF54	Dd2
NF54	Sensitive	237	1.0	
K1	Decreased sensitivity to chloroquine	278	1.2	
7G8	Decreased sensitivity to chloroquine	293	1.2	
TM90C2B	Decreased sensitivity to atovaquone	194	0.8	
Cam3.1	Decreased sensitivity to artemisinin	244	1.0	
Dd2	Decreased sensitivity to chloroquine	220	0.9	1.0
Dd2 DDD107498	Decreased sensitivity to eEF2 inhibitor, DDD107498	211		1.0
Dd2	Decreased sensitivity to PI4K inhibitor, MMV390048	164		0.7
MMV390048				
Dd2 DSM265	Decreased sensitivity to DHODH inhibitor, DSM265	183		0.8
Dd2 GNF156	Decreased sensitivity to GNF156	195		0.9
Dd2 ELQ300	Decreased sensitivity to cyt bc1 inhibitor, ELQ300	174		0.8

^a 72-hour [³H]-hypoxanthine incorporation assay. ^b mean values from 2 biological replicates.

Supplemental Table S3

Related to Fig. 1 [↗](#).

MIPS2673 effectiveness against a panel of drug resistant *P. falciparum* strains^a [↗](#).

Compound	EC ₅₀ (nM) ^a		
	Early-stage Gametocytes	Late-stage Gametocytes	Mature-stage Gametocytes
MIPS2673	1660 ± 71	5300 ± 1923	98% (120 μM) ^b
Artesunate	16 ± 5.0	26 ± 2.0	51 ± 23

^a EC₅₀ values were determined in technical duplicate from 2-3 biological repeats.

^b % inhibition at 120 μM (EC₅₀ not determined).

Supplemental Table S4

Related to Fig. 1 [↗](#).

MIPS2673 effectiveness against *P. falciparum* early (I-III)-, late (IV-V)-, and mature (V) gametocyte stages of NF54-pfs16-GFP parasites.

	M1 MIPS2673
PDB ID	8SLO
<i>Data collection statistics</i>	
Wavelength	0.953700
Resolution range	41.4 - 1.55 (1.605 - 1.55)
Space group	P 21 21 21
Unit cell	76.4 108.5 117.6 90 90 90
Total reflections	1038039 (39097)
Unique reflections	139690 (12944)
Multiplicity	7.4 (6.4)
Completeness (%)	98.39 (91.92)
Mean I/sigma(I)	8.5 (0.9)
Wilson B-factor	17.19
R-pim (all I+ & I-)	0.057 (0.772)
CC1/2	0.997 (0.385)
<i>Data Refinement statistics</i>	
Reflections used in refinement	139628 (12905)
Reflections used for R-free	7039 (646)
R-work	0.1712 (0.2886)
R-free	0.2021 (0.3072)
Number of non-hydrogen atoms	8774
macromolecules	7432
ligands	45
solvent	1297
Protein residues	889
RMS(bonds)	0.007
RMS(angles)	0.88
Ramachandran favored (%)	98.20
Ramachandran allowed (%)	1.69
Ramachandran outliers (%)	0.11
Rotamer outliers (%)	0.24
Clashscore	2.35
Average B-factor	21.92
macromolecules	20.04
ligands	19.42
solvent	32.76

Statistics for the highest-resolution shell are shown in parentheses.

Supplemental Table S5

Related to Fig. 1 .

Data collection and refinement statistics.

References

1. WHO (2022) **World Malaria Report 2022**
2. Price R.N., Commons R.J., Battle K.E., Thriemer K., Mendis K. (2020) **Plasmodium vivax in the Era of the Shrinking P. falciparum Map** *Trends Parasitol* **36**:560–570 <https://doi.org/10.1016/j.pt.2020.03.009>
3. Phillips M.A., Burrows J.N., Manyando C., van Huijsduijnen R.H., Van Voorhis W.C., Wells T.N.C. (2017) **Malaria** *Nat. Rev. Dis. Primers* **3** <https://doi.org/10.1038/nrdp.2017.50>
4. van der Pluijm R.W., Imwong M., Chau N.H., Hoa N.T., Thuy-Nhien N.T., Thanh N.V., Jittamala P., Hanboonkunupakarn B., Chutasmit K., Saelow C. (2019) **Determinants of dihydroartemisinin-piperaquine treatment failure in Plasmodium falciparum malaria in Cambodia, Thailand, and Vietnam: a prospective clinical, pharmacological, and genetic study** *Lancet Infect. Dis* **19**:952–961 [https://doi.org/10.1016/S1473-3099\(19\)30391-3](https://doi.org/10.1016/S1473-3099(19)30391-3)
5. Balikagala B., Fukuda N., Ikeda M., Katuro O.T., Tachibana S.-I., Yamauchi M., Opio W., Emoto S., Anywar D.A., Kimura E. (2021) **Evidence of artemisinin-resistant malaria in Africa** *N. Engl. J. Med* **385**:1163–1171 <https://doi.org/10.1056/NEJMoa2101746>
6. Liu J., Istvan E.S., Gluzman I.Y., Gross J., Goldberg D.E. (2006) **Plasmodium falciparum ensures its amino acid supply with multiple acquisition pathways and redundant proteolytic enzyme systems** *Proc. Natl. Acad. Sci. U. S. A* **103**:8840–8845 <https://doi.org/10.1073/pnas.0601876103>
7. Lew V.L., Tiffert T., Ginsburg H. (2003) **Excess hemoglobin digestion and the osmotic stability of Plasmodium falciparum-infected red blood cells** *Blood* **101**:4189–4194 <https://doi.org/10.1182/blood-2002-08-2654>
8. Krugliak M., Zhang J., Ginsburg H. (2002) **Intraerythrocytic Plasmodium falciparum utilizes only a fraction of the amino acids derived from the digestion of host cell cytosol for the biosynthesis of its proteins** *Mol. Biochem. Parasitol* **119**:249–256 [https://doi.org/10.1016/S0166-6851\(01\)00427-3](https://doi.org/10.1016/S0166-6851(01)00427-3)
9. Loria P., Miller S., Foley M., Tilley L. (1999) **Inhibition of the peroxidative degradation of haem as the basis of action of chloroquine and other quinoline antimalarials** *Biochem. J* **339**:363–370
10. Goldberg D.E. *et al.* (2005) **Hemoglobin degradation** *Malaria: Drugs, disease and post-genomic biology* :275–291 https://doi.org/10.1007/3-540-29088-5_11
11. Edgar R.C.S., Siddiqui G., Hjerrild K., Malcolm T.R., Vinh N.B., Webb C.T., Holmes C., MacRaild C.A., Chernih H.C., Suen W.W. (2022) **Genetic and chemical validation of Plasmodium falciparum aminopeptidase PfA-M17 as a drug target in the hemoglobin digestion pathway** *eLife* **11** <https://doi.org/10.7554/eLife.80813>
12. Dalal S., Klemmba M. (2007) **Roles for two aminopeptidases in vacuolar hemoglobin catabolism in Plasmodium falciparum** *J. Biol. Chem* **282**:35978–35987 <https://doi.org/10.1074/jbc.M703643200>

13. Skinner-Adams T.S., Lowther J., Teuscher F., Stack C.M., Grembecka J., Mucha A., Kafarski P., Trenholme K.R., Dalton J.P., Gardiner D.L. (2007) **Identification of Phosphinate Dipeptide Analog Inhibitors Directed against the Plasmodium falciparum M17 Leucine Aminopeptidase as Lead Antimalarial Compounds** *J. Med. Chem* **50**:6024–6031 <https://doi.org/10.1021/jm070733v>
14. Skinner-Adams T.S., Peatey C.L., Anderson K., Trenholme K.R., Krige D., Brown C.L., Stack C., Nsangou D.M.M., Mathews R.T., Thivierge K. (2012) **The Aminopeptidase Inhibitor CHR-2863 Is an Orally Bioavailable Inhibitor of Murine Malaria** *Antimicrob. Agents Chemother* **56**:3244–3249 <https://doi.org/10.1128/AAC.06245-11>
15. Harbut M.B., Velmourougane G., Dalal S., Reiss G., Whisstock J.C., Onder O., Brisson D., McGowan S., Klemba M., Greenbaum D.C. (2011) **Bestatin-based chemical biology strategy reveals distinct roles for malaria M1- and M17-family aminopeptidases** *Proc. Natl. Acad. Sci. U. S. A* **108**:E526–534 <https://doi.org/10.1073/pnas.1105601108>
16. McGowan S. (2013) **Working in concert: the metalloaminopeptidases from Plasmodium falciparum** *Curr. Opin. Struct. Biol* **23**:828–835 <https://doi.org/10.1016/j.sbi.2013.07.015>
17. Nankya-Kitaka M.F., Curley G.P., Gavigan C.S., Bell A., Dalton J.P. (1998) **Plasmodium chabaudi chabaudi and P. falciparum: inhibition of aminopeptidase and parasite growth by bestatin and nitrobestatin** *Parasitol. Res* **84**:552–558 <https://doi.org/10.1007/s004360050447>
18. Bounaadja L., Schmitt M., Albrecht S., Mouray E., Tarnus C., Florent I. (2017) **Selective inhibition of PfA-M1, over PfA-M17, by an amino-benzosuberone derivative blocks malaria parasites development in vitro and in vivo** *Malar. J* **16** <https://doi.org/10.1186/s12936-017-2032-4>
19. Deprez-Poulain R., Flipo M., Piveteau C., Leroux F., Dassonneville S., Florent I., Maes L., Cos P., Deprez B. (2012) **Structure–Activity Relationships and Blood Distribution of Antiplasmodial Aminopeptidase-1 Inhibitors** *J. Med. Chem* **55**:10909–10917 <https://doi.org/10.1021/jm301506h>
20. Drinkwater N., Vinh N.B., Mistry S.N., Bamert R.S., Ruggeri C., Holleran J.P., Loganathan S., Paiardini A., Charman S.A., Powell A.K. (2016) **Potent dual inhibitors of Plasmodium falciparum M1 and M17 aminopeptidases through optimization of S1 pocket interactions** *European Journal of Medicinal Chemistry* **110**:43–64 <https://doi.org/10.1016/j.ejmech.2016.01.015>
21. Vinh N.B., Drinkwater N., Malcolm T.R., Kassiou M., Lucantoni L., Grin P.M., Butler G.S., Duffy S., Overall C.M., Avery V.M. (2019) **Hydroxamic Acid Inhibitors Provide Cross-Species Inhibition of Plasmodium M1 and M17 Aminopeptidases** *J. Med. Chem* **62**:622–640 <https://doi.org/10.1021/acs.jmedchem.8b01310>
22. Gonzalez-Bacerio J., Fando R., Monte-Martinez d.A., Charli J.-L., Chávez d.I.A.M. (2014) **Plasmodium falciparum M1-Aminopeptidase: A Promising Target for the Development of Antimalarials** *Current Drug Targets* **15**:1144–1165 <https://doi.org/10.2174/1389450115666141024115641>
23. Mistry S.N. *et al.* (2014) **Two-Pronged Attack: Dual Inhibition of Plasmodium falciparum M1 and M17 Metalloaminopeptidases by a Novel Series of Hydroxamic Acid-Based Inhibitors** *J. Med. Chem* **57**:9168–9183 <https://doi.org/10.1021/jm501323a>

24. Ruggeri C., Drinkwater N., Sivaraman K.K., Bamert R.S., McGowan S., Paiardini A. (2015) **Identification and Validation of a Potent Dual Inhibitor of the *P. falciparum* M1 and M17 Aminopeptidases Using Virtual Screening** *PLoS One* **10** <https://doi.org/10.1371/journal.pone.0138957>
25. Kannan Sivaraman K., Paiardini A., Sieńczyk M., Ruggeri C., Oellig C.A., Dalton J.P., Scammells P.J., Drag M., McGowan S. (2013) **Synthesis and Structure–Activity Relationships of Phosphonic Arginine Mimetics as Inhibitors of the M1 and M17 Aminopeptidases from *Plasmodium falciparum*** *J. Med. Chem* **56**:5213–5217 <https://doi.org/10.1021/jm4005972>
26. Allary M., Schrevel J., Florent I. (2002) **Properties, stage-dependent expression and localization of *Plasmodium falciparum* M1 family zinc-aminopeptidase** *Parasitology* **125**:1–10 <https://doi.org/10.1017/S0031182002001828>
27. Azimzadeh O., Sow C., Gèze M., Nyalwidhe J., Florent I. (2010) ***Plasmodium falciparum* PfA-M1 aminopeptidase is traicked via the parasitophorous vacuole and marginally delivered to the food vacuole** *Malar. J* **9** <https://doi.org/10.1186/1475-2875-9-189>
28. Mathew R., Wunderlich J., Thivierge K., Cwiklinski K., Dumont C., Tilley L., Rohrbach P., Dalton J.P. (2021) **Biochemical and cellular characterisation of the *Plasmodium falciparum* M1 alanyl aminopeptidase (PfM1AAP) and M17 leucyl aminopeptidase (PfM17LAP)** *Sci. Rep* **11** <https://doi.org/10.1038/s41598-021-82499-4>
29. Poreba M., McGowan S., Skinner-Adams T.S., Trenholme K.R., Gardiner D.L., Whisstock J.C., To J., Salvesen G.S., Dalton J.P., Drag M. (2012) **Fingerprinting the Substrate Specificity of M1 and M17 Aminopeptidases of Human Malaria, *Plasmodium falciparum*** *PLoS One* **7** <https://doi.org/10.1371/journal.pone.0031938>
30. McGowan S., Porter C.J., Lowther J., Stack C.M., Golding S.J., Skinner-Adams T.S., Trenholme K.R., Teuscher F., Donnelly S.M., Grembecka J. (2009) **Structural basis for the inhibition of the essential *Plasmodium falciparum* M1 neutral aminopeptidase** *Proceedings of the National Academy of Sciences* **106**:2537–2542 <https://doi.org/10.1073/pnas.0807398106>
31. Flipo M., Beghyn T., Leroux V., Florent I., Deprez B.P., Deprez-Poulain R.F. (2007) **Novel Selective Inhibitors of the Zinc Plasmodial Aminopeptidase PfA-M1 as Potential Antimalarial Agents** *J. Med. Chem* **50**:1322–1334 <https://doi.org/10.1021/jm061169b>
32. Harbut M.B., Velmourougane G., Reiss G., Chandramohanadas R., Greenbaum D.C. (2008) **Development of bestatin-based activity-based probes for metallo-aminopeptidases** *Bioorg. Med. Chem. Lett* **18**:5932–5936 <https://doi.org/10.1016/j.bmcl.2008.09.021>
33. Velmourougane G., Harbut M.B., Dalal S., McGowan S., Oellig C.A., Meinhardt N., Whisstock J.C., Klemba M., Greenbaum D.C. (2011) **Synthesis of New (–)-Bestatin-Based Inhibitor Libraries Reveals a Novel Binding Mode in the S1 Pocket of the Essential Malaria M1 Metalloaminopeptidase** *J. Med. Chem* **54**:1655–1666 <https://doi.org/10.1021/jm101227t>
34. McGowan S., Oellig C.A., Birru W.A., Caradoc-Davies T.T., Stack C.M., Lowther J., Skinner-Adams T., Mucha A., Kafarski P., Grembecka J. (2010) **Structure of the *Plasmodium falciparum* M17 aminopeptidase and significance for the design of drugs targeting the neutral exopeptidases** *Proceedings of the National Academy of Sciences* **107**:2449–2454 <https://doi.org/10.1073/pnas.0911813107>

35. Challis M.P., Devine S.M., Creek D.J. (2022) **Current and emerging target identification methods for novel antimalarials** *Int. J. Parasitol. Drugs Drug Resist* **20**:135–144 <https://doi.org/10.1016/j.ijpddr.2022.11.001>
36. Bailey B.L., Nguyen W., Cowman A.F., Sleebs B.E. (2023) **Chemo-proteomics in antimalarial target identification and engagement** <https://doi.org/10.1002/med.21975>
37. Carolino K., Winzeler E.A. (2020) **The antimalarial resistome – finding new drug targets and their modes of action** *Current Opinion in Microbiology* **57**:49–55 <https://doi.org/10.1016/j.mib.2020.06.004>
38. Baragana B., Hallyburton I., Lee M.C.S., Norcross N.R., Grimaldi R., Otto T.D., Proto W.R., Blagborough A.M., Meister S., Wirjanata G. (2015) **A novel multiple-stage antimalarial agent that inhibits protein synthesis** *Nature* **522**:315–320 <https://doi.org/10.1038/nature14451>
39. Summers R.L., Pasaje C.F.A., Pisco J.P., Striepen J., Luth M.R., Kumpornsin K., Carpenter E.F., Munro J.T., Lin D., Plater A. (2021) **Chemogenomics identifies acetyl-coenzyme A synthetase as a target for malaria treatment and prevention** <https://doi.org/10.1016/j.chembiol.2021.07.010>
40. Cowell A.N., Istvan E.S., Lukens A.K., Gomez-Lorenzo M.G., Vanaerschot M., Sakata-Kato T., Flannery E.L., Magistrado P., Owen E., Abraham M. (2018) **Mapping the malaria parasite druggable genome by using in vitro evolution and chemogenomics** *Science* **359**:191–199 <https://doi.org/10.1126/science.aan4472>
41. Flannery E.L., McNamara C.W., Kim S.W., Kato T.S., Li F., Teng C.H., Gagaring K., Manary M.J., Barboa R., Meister S. (2015) **Mutations in the P-Type Cation-Transporter ATPase 4, PfATP4, Mediate Resistance to Both Aminopyrazole and Spiroindolone Antimalarials** *ACS Chem. Biol* **10**:413–420 <https://doi.org/10.1021/cb500616x>
42. Yang T., Otilie S., Istvan E.S., Godinez-Macias K.P., Lukens A.K., Baragaña B., Campo B., Walpole C., Niles J.C., Chibale K. (2021) **MalDA, Accelerating Malaria Drug Discovery** *Trends Parasitol* **37**:493–507 <https://doi.org/10.1016/j.pt.2021.01.009>
43. Siddiqui G., Giannangelo C., De Paoli A., Schuh A.K., Heimsch K.C., Anderson D., Brown T.G., MacRaid C.A., Wu J., Wang X. (2022) **Peroxide antimalarial drugs target redox homeostasis in Plasmodium falciparum infected red blood cells** <https://doi.org/10.1021/acscinfecdis.1c00550>
44. Molina D.M., Jafari R., Ignatushchenko M., Seki T., Larsson E.A., Dan C., Sreekumar L., Cao Y., Nordlund P. (2013) **Monitoring drug target engagement in cells and tissues using the cellular thermal shift assay** *Science* **341**:84–87 <https://doi.org/10.1126/science.1233606>
45. Dziekan J.M., Yu H., Chen D., Dai L., Wirjanata G., Larsson A., Prabhu N., Sobota R.M., Bozdech Z., Nordlund P. (2019) **Identifying purine nucleoside phosphorylase as the target of quinine using cellular thermal shift assay** *Sci. Transl. Med* **11** <https://doi.org/10.1126/scitranslmed.aau3174>
46. Piazza I., Beaton N., Bruderer R., Knobloch T., Barbisan C., Chandat L., Sudau A., Siepe I., Rinner O., de Souza N. (2020) **A machine learning-based chemoproteomic approach to identify drug targets and binding sites in complex proteomes** *Nat. Commun* **11** <https://doi.org/10.1038/s41467-020-18071-x>

47. Wirjanata G., Dziekan J.M., Lin J., Sahili A.E., Binte Zulkifli N.E., Boentoro J., Go K.D., Yu H., Partridge A., Olsen D. (2021) **Identification of an inhibitory pocket in falcilysin bound by chloroquine provides a new avenue for malaria drug development** <https://doi.org/10.1101/2021.04.08.438947>
48. Gavigan C.S., Dalton J.P., Bell A. (2001) **The role of aminopeptidases in haemoglobin degradation in Plasmodium falciparum-infected erythrocytes** *Molecular and biochemical parasitology* **117**:37–48 [https://doi.org/10.1016/S0166-6851\(01\)00327-9](https://doi.org/10.1016/S0166-6851(01)00327-9)
49. Yang T., Xie S.C., Cao P., Giannangelo C., McCaw J., Creek D.J., Charman S.A., Klonis N., Tilley L. (2016) **Comparison of the exposure time-dependence of the activities of synthetic ozonide antimalarials and dihydroartemisinin against K13 wild-type and mutant Plasmodium falciparum strains** *Antimicrob. Agents Chemother* **60**:4501–4510 <https://doi.org/10.1128/AAC.00574-16>
50. Zhang M., Wang C., Otto T.D., Oberstaller J., Liao X., Adapa S.R., Udenze K., Bronner I.F., Casandra D., Mayho M. (2018) **Uncovering the essential genes of the human malaria parasite Plasmodium falciparum by saturation mutagenesis** *Science* **360** <https://doi.org/10.1126/science.aap7847>
51. Siddiqui G., De Paoli A., MacRaid C.A., Sexton A.E., Boulet C., Shah A.D., Batty M.B., Schittenhelm R.B., Carvalho T.G., Creek D.J. (2022) **A new mass spectral library for high-coverage and reproducible analysis of the Plasmodium falciparum-infected red blood cell proteome** *GigaScience* **11** <https://doi.org/10.1093/gigascience/giac008>
52. George A.L., Sidgwick F.R., Watt J.E., Martin M.P., Trost M., Marín-Rubio J.L., Dueñas M.E. (2023) **Comparison of Quantitative Mass Spectrometric Methods for Drug Target Identification by Thermal Proteome Profiling** <https://doi.org/10.1021/acs.jproteome.3c00111>
53. Dziekan J.M., Wirjanata G., Dai L., Go K.D., Yu H., Lim Y.T., Chen L., Wang L.C., Puspita B., Prabhu N. (2020) **Cellular thermal shift assay for the identification of drug–target interactions in the Plasmodium falciparum proteome** *Nat. Protoc* **15**:1881–1921 <https://doi.org/10.1038/s41596-020-0310-z>
54. Piazza I., Kochanowski K., Cappelletti V., Fuhrer T., Noor E., Sauer U., Picotti P. (2018) **A Map of Protein-Metabolite Interactions Reveals Principles of Chemical Communication** *Cell* **172**:358–372 <https://doi.org/10.1016/j.cell.2017.12.006>
55. Creek D.J., Chua H.H., Cobbold S.A., Nijagal B., MacRae J.I., Dickerman B.K., Gilson P.R., Ralph S.A., McConville M.J. (2016) **Metabolomics-based screening of the malaria box reveals both novel and established mechanisms of action** *Antimicrob. Agents Chemother* **60**:6650–6663 <https://doi.org/10.1128/aac.01226-16>
56. Giannangelo C., Siddiqui G., De Paoli A., Anderson B.M., Edgington-Mitchell L.E., Charman S.A., Creek D.J. (2020) **System-wide biochemical analysis reveals ozonide antimalarials initially act by disrupting Plasmodium falciparum haemoglobin digestion** *PLoS Pathog* **16** <https://doi.org/10.1371/journal.ppat.1008485>
57. Birrell G.W., Challis M.P., De Paoli A., Anderson D., Devine S.M., Heffernan G.D., Jacobus D.P., Edstein M.D., Siddiqui G., Creek D.J. (2020) **Multi-omic characterization of the mode of action of a potent new antimalarial compound, JPC-3210, Against Plasmodium falciparum** *Mol. Cell. Proteomics* **19**:308–325 <https://doi.org/10.1074/mcp.RA119.001797>

58. Milne R., Wiedemar N., Corpas-Lopez V., Moynihan E., Wall R.J., Dawson A., Robinson D.A., Shepherd S.M., Smith R.J., Hallyburton I. (2022) **Toolkit of Approaches To Support Target-Focused Drug Discovery for Plasmodium falciparum Lysyl tRNA Synthetase** *ACS Infect. Dis* **8**:1962–1974 <https://doi.org/10.1021/acsinfecdis.2c00364>
59. Gillet L.C., Navarro P., Tate S., Röst H., Selevsek N., Reiter L., Bonner R., Aebersold R. (2012) **Targeted Data Extraction of the MS/MS Spectra Generated by Data-independent Acquisition: A New Concept for Consistent and Accurate Proteome Analysis** *Mol. Cell. Proteomics* **11** <https://doi.org/10.1074/mcp.O111.016717>
60. Papakyriakou A., Stratikos E. (2017) **The role of conformational dynamics in antigen trimming by intracellular aminopeptidases** *Front. Immunol* **8** <https://doi.org/10.3389/fimmu.2017.00946>
61. Moore D.S., Brines C., Jewhurst H., Dalton J.P., Tikhonova I.G. (2018) **Steered molecular dynamics simulations reveal critical residues for (un)binding of substrates, inhibitors and a product to the malarial M1 aminopeptidase** *PLoS Comput. Biol* **14** <https://doi.org/10.1371/journal.pcbi.1006525>
62. Shang X., Wang C., Fan Y., Guo G., Wang F., Zhao Y., Sheng F., Tang J., He X., Yu X. (2022) **Genome-wide landscape of ApiAP2 transcription factors reveals a heterochromatin-associated regulatory network during Plasmodium falciparum blood-stage development** *Nucleic Acids Res* **50**:3413–3431 <https://doi.org/10.1093/nar/gkac176>
63. Ragheb D., Dalal S., Bompiani K.M., Ray W.K., Klemba M. (2011) **Distribution and biochemical properties of an M1-family aminopeptidase in Plasmodium falciparum indicate a role in vacuolar hemoglobin catabolism** *J. Biol. Chem* **286**:27255–27265 <https://doi.org/10.1074/jbc.M111.225318>
64. Plouffe D., Brinker A., McNamara C., Henson K., Kato N., Kuhen K., Nagle A., Adrian F., Matzen J.T., Anderson P. (2008) **In silico activity profiling reveals the mechanism of action of antimalarials discovered in a high-throughput screen** *Proceedings of the National Academy of Sciences of the United States of America* **105**:9059–9064 <https://doi.org/10.1073/pnas.0802982105>
65. Gamo F.J., Sanz L.M., Vidal J., de Cozar C., Alvarez E., Lavandera J.L., Vanderwall D.E., Green D.V., Kumar V., Hasan S. (2010) **Thousands of chemical starting points for antimalarial lead identification** *Nature* **465**:305–310 <https://doi.org/10.1038/nature09107>
66. Guiguemde W.A., Shelat A.A., Garcia-Bustos J.F., Diagana T.T., Gamo F.J., Guy R.K. (2012) **Global phenotypic screening for antimalarials** *Chem Biol* **19**:116–129 <https://doi.org/10.1016/j.chembiol.2012.01.004>
67. Trager W., Jensen J.B. (1976) **Human malaria parasites in continuous culture** *Science* **193**:673–675 <https://doi.org/10.1126/science.781840>
68. Lambros C., Vanderberg J.P. (1979) **Synchronization of Plasmodium falciparum erythrocytic stages in culture** *J. Parasitol* **65**:418–420 <https://doi.org/10.2307/3280287>
69. Rappsilber J., Ishihama Y., Mann M. (2003) **Stop and go extraction tips for matrix-assisted laser desorption/ionization, nanoelectrospray, and LC/MS sample pretreatment in proteomics** *Anal. Chem* **75**:663–670 <https://doi.org/10.1021/ac026117i>

70. Alexa A., Rahnenführer J. (2020) **topGO: Enrichment Analysis for Gene Ontology, R package version 2.42.0**
71. Alexa A., Rahnenführer J., Lengauer T. (2006) **Improved scoring of functional groups from gene expression data by decorrelating GO graph structure** *Bioinformatics* **22**:1600–1607 <https://doi.org/10.1093/bioinformatics/btl140>
72. Malcolm T.R., Swiderska K.W., Hayes B.K., Webb C.T., Drag M., Drinkwater N., McGowan S. (2021) **Mapping the substrate specificity of the Plasmodium M1 and M17 aminopeptidases** *Biochem. J* **478**:2697–2713 <https://doi.org/10.1042/bcj20210172>
73. Drinkwater N., Sivaraman K.K., Bamert R.S., Rut W., Mohamed K., Vinh N.B., Scammells P.J., Drag M., McGowan S. (2016) **Structure and substrate fingerprint of aminopeptidase P from Plasmodium falciparum** *Biochem. J* **473**:3189–3204 <https://doi.org/10.1042/bcj20160550>
74. Sivaraman K.K., Oellig C.A., Huynh K., Atkinson S.C., Poreba M., Perugini M.A., Trenholme K.R., Gardiner D.L., Salvesen G., Drag M. (2012) **X-ray crystal structure and specificity of the Plasmodium falciparum malaria aminopeptidase PfM18AAP** *J. Mol. Biol* **422**:495–507 <https://doi.org/10.1016/j.jmb.2012.06.006>
75. Cowieson N.P., Aragao D., Clift M., Ericsson D.J., Gee C., Harrop S.J., Mudie N., Panjikar S., Price J.R., Riboldi-Tunncliffe A. (2015) **MX1: a bending-magnet crystallography beamline serving both chemical and macromolecular crystallography communities at the Australian Synchrotron** *Journal of Synchrotron Radiation* **22**:187–190 <https://doi.org/10.1107/S1600577514021717>
76. Kabsch W. (2010) **XDS** *Acta Crystallogr. D Biol. Crystallogr* **66**:125–132 <https://doi.org/10.1107/S0907444909047337>
77. Evans P.R., Murshudov G.N. (2013) **How good are my data and what is the resolution?** *Acta Crystallographica Section D* **69**:1204–1214 <https://doi.org/10.1107/S0907444913000061>
78. Winn M.D., Ballard C.C., Cowtan K.D., Dodson E.J., Emsley P., Evans P.R., Keegan R.M., Krissinel E.B., Leslie A.G., McCoy A. (2011) **Overview of the CCP4 suite and current developments** *Acta Crystallogr. D Biol. Crystallogr* **67**:235–242 <https://doi.org/10.1107/S0907444910045749>
79. McCoy A.J., Grosse-Kunstleve R.W., Adams P.D., Winn M.D., Storoni L.C., Read R.J. (2007) **Phaser crystallographic software** *J Appl Crystallogr* **40**:658–674 <https://doi.org/10.1107/S0021889807021206>
80. Adams P.D., Afonine P.V., Bunkóczi G., Chen V.B., Davis I.W., Echols N., Headd J.J., Hung L.W., Kapral G.J., Grosse-Kunstleve R.W. (2010) **PHENIX: a comprehensive Python-based system for macromolecular structure solution** *Acta Crystallogr. D Biol. Crystallogr* **66**:213–221 <https://doi.org/10.1107/S0907444909052925>
81. Emsley P., Cowtan K. (2004) **Coot: model-building tools for molecular graphics** *Acta Crystallogr. D Biol. Crystallogr* **60**:2126–2132 <https://doi.org/10.1107/S0907444904019158>
82. Emsley P., Lohkamp B., Scott W.G., Cowtan K. (2010) **Features and development of Coot** *Acta Crystallogr. D Biol. Crystallogr* **66**:486–501 <https://doi.org/10.1107/S0907444910007493>
83. Smilkstein M., Sriwilaijaroen N., Kelly J.X., Wilairat P., Riscoe M. (2004) **Simple and inexpensive fluorescence-based technique for high-throughput antimalarial drug screening** *Antimicrob. Agents Chemother* **48**:1803–1806 <https://doi.org/10.1128/AAC.48.5.1803-1806.2004>

84. Snyder C., Chollet J., Santo-Tomas J., Scheurer C., Wittlin S. (2007) **In vitro and in vivo interaction of synthetic peroxide RBx11160 (OZ277) with piperazine in Plasmodium models** *Exp. Parasitol* **115**:296–300 <https://doi.org/10.1016/j.exppara.2006.09.016>
85. Duffy S., Avery V.M. (2013) **Identification of inhibitors of Plasmodium falciparum gametocyte development** *Malar. J* **12** <https://doi.org/10.1186/1475-2875-12-408>
86. Duffy S., Loganathan S., Holleran J.P., Avery V.M. (2016) **Large-scale production of Plasmodium falciparum gametocytes for malaria drug discovery** *Nat. Protoc* **11**:976–992 <https://doi.org/10.1038/nprot.2016.056>
87. Lucantoni L., Loganathan S., Avery V.M. (2017) **The need to compare: assessing the level of agreement of three high-throughput assays against Plasmodium falciparum mature gametocytes** *Sci. Rep* **7** <https://doi.org/10.1038/srep45992>
88. Duffy S., Avery V.M. (2012) **Development and Optimization of a Novel 384-Well Anti-Malarial Imaging Assay Validated for High-Throughput Screening** *The American Society of Tropical Medicine and Hygiene* **86**:84–92 <https://doi.org/10.4269/ajtmh.2012.11-0302>
89. Bevan C.D., Lloyd R.S. (2000) **A high-throughput screening method for the determination of aqueous drug solubility using laser nephelometry in microtiter plates** *Anal. Chem* **72**:1781–1787 <https://doi.org/10.1021/ac9912247>
90. Perez-Riverol Y., Bai J., Bandla C., García-Seisdedos D., Hewapathirana S., Kamatchinathan S., Kundu Deepti J., Prakash A., Frericks-Zipper A., Eisenacher M. (2021) **The PRIDE database resources in 2022: a hub for mass spectrometry-based proteomics evidences** *Nucleic Acids Res* **50**:D543–D552 <https://doi.org/10.1093/nar/gkab1038>

Article and author information

Darren Creek

Monash University

For correspondence: darren.creek@monash.edu

ORCID ID: [0000-0001-7497-7082](https://orcid.org/0000-0001-7497-7082)

Carlo Giannangelo

Monash University

Matthew Challis

Monash University

Ghizal Siddiqui

Monash University

Rebecca Edgar

Deakin University

Tess Malcolm

Monash University

Chaille Webb

Monash University

Nyssa Drinkwater

Monash University

Natalie Vinh

Monash University

Christopher MacRaid

Monash University

Natalie Counihan

Deakin University

Sandra Duffy

Griffith University

ORCID iD: [0000-0002-2814-1193](https://orcid.org/0000-0002-2814-1193)

Sergio Wittlin

University of Basel

Shane Devine

Walter and Eliza Hall Institute

ORCID iD: [0000-0001-8727-7392](https://orcid.org/0000-0001-8727-7392)

Vicky Avery

Griffith University

Tania de Koning-Ward

Deakin University

Peter Scammells

Monash Institute of Pharmaceutical Sciences

Sheena McGowan

Monash University

ORCID iD: [0000-0001-6863-1106](https://orcid.org/0000-0001-6863-1106)

Copyright

© 2024, Creek et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Dominique Soldati-Favre

University of Geneva, Geneva, Switzerland

Senior Editor

Dominique Soldati-Favre

University of Geneva, Geneva, Switzerland

Reviewer #1 (Public Review):

Summary:

The article "Chemoproteomics validates selective targeting of Plasmodium M1 alanyl aminopeptidase as a cross-species strategy to treat malaria" presents a series of biochemical methods based on proteomics and metabolomics, as a means to:

(1) validate the specific targeting of biologically active molecules (MIPS2673) towards a defined (unique) protein target within a parasite

and

(2) to explore whether by quantifying the perturbations generated at the level of the parasite metabolome, it is possible to extrapolate which metabolic pathway has been disrupted by using this biologically active molecule and whether this may further confirm selective targeting in parasites of the expected (or in-vitro targeted) enzyme (here PfA-1).

The inhibitor used in this work by the authors (MIPS2673) is to my knowledge a novel one, although belonging to a chemical series previously explored by the authors, which recently enabled them to discover a specific PfA-M17 inhibitor, MIPS2571 (Edgard et al., 2022, ref 11 of this current work). Indeed, inhibitors specifically targeting either PfA-M1 or PfA-M17 (and not both, as currently done in the past) are scarce today, and highly needed to functionally characterize these two zinc-aminopeptidases. MIPS2673, blocks the development of erythrocytic stages of *Plasmodium falciparum* with an EC₅₀ of 324 nM, blocks the parasite development at the young trophozoite stage at 5x EC₅₀ (but at ring stages at 10xEC₅₀, figure 1E), and inhibits the enzymatic activity of PfA-M1 (and its ortholog Pv-M1) but not of the related malarial metallo-aminopeptidases (M17 and M18 families) nor the human metalloenzymes from closely related enzymatic families, supporting its selective targeting of PfA-M1 (and Pv-M1).

All experiments are carried out in vitro (e.g. biochemical studies such as enzymology, proteomics, metabolomics) and on cultured parasites (erythrocyte stages of *Plasmodium falciparum* and several gametocytes stages obtained in vitro); there are no in vivo manipulations. The work related to *Plasmodium vivax*, which justifies the "cross-species" indication in the title of the article, is restricted to using a recombinant form of the M1-family aminopeptidase in enzymatic assays. The rest of the work concerns only *Plasmodium falciparum*. While I found globally that this work is original and brings new data and above all proposes chemical validation approaches that could be used for other target validations under similar limiting conditions (impossibility of KO of the gene), I have some specific questions to address to the authors.

Strengths and weaknesses:

-The chemoproteomic approach, that explores the ability of MIPS2673 to more significantly "protect" the putative target (PfA-M1) against thermal degradation or enzymatic attack (by proteinase K), to document its selective targeting towards PfA-M1 (the inhibitor, once associated with its target, is expected to stabilize its structure or prevent the action of end proteases), uses several concentrations of MIPS2673 and provides convincing results. My main criticism is that these tests are carried out with parasite extracts enriched in 30-38 hours old forms, and restricted to the fraction of soluble proteins isolated from these

parasitic forms, which still limits the scope of the analysis. It is clear that this methodological approach is a choice that can be argued both biologically (PfA-M1 is well expressed in these stages of the parasite development) and biochemically (it is difficult to do proteomic analyses on insoluble proteins) but I regret that the authors do not discuss these limitations further, notably, I would have expected (from Figure 1E) some targets to be also present at ring stages.

-The metabolomic approach, by documenting the ability of MIPS2673 to selectively increase the number of non-hydrolyzed dipeptides in treated versus untreated parasites is another argument in favor of the selective targeting of PfA-M1 by MIPS2673, in particular by its broad-spectrum aminopeptidase action preferentially targeting peptides resulting from the degradation of hemoglobin by the parasite. The relative contribution of peptides derived from host hemoglobin versus other parasite proteins is, however, little discussed.

The work as a whole remains highly interesting, both for the specific topic of PfA-M1's role in parasite biology and for the method, applicable to other malarial drug contexts.

<https://doi.org/10.7554/eLife.92990.1.sa2>

Reviewer #2 (Public Review):

In this manuscript, the authors first developed a new small molecular inhibitor that could target specifically the M1 metalloproteases of both important malaria parasite species *Plasmodium falciparum* and *P. vivax*. This was done by a chemical modification of a previously developed molecule that targets PfM1 as well as PfM17 and possibly other Plasmodial metalloproteases. After the successful chemical synthesis, the authors showed that the derived inhibitor, named MIPS2673, has a strong antiparasitic activity with IC₅₀ 342 nM and it is highly specific for M1. With this in mind, the authors first carried out two large-scale proteomics to confirm the MIPS2673 interaction with PfM1 in the context of the total *P. falciparum* protein lysate. This was done first by using thermal shift profiling and subsequently limited proteolysis. While the first demonstrated overall interaction, the latter (limited proteolysis) could map more specifically the site of MIPS2673-PfM1 interaction, presumably the active site. Subsequent metabolomics analysis showed that MIPS2673 cytotoxic inhibitory effect leads to the accumulation of short peptides many of which originate from hemoglobin. Based on that the authors argue that the MIPS2673 mode of action (MOA) involves inhibition of hemoglobin digestion that in turn inhibits the parasite growth and development.

<https://doi.org/10.7554/eLife.92990.1.sa1>

Reviewer #3 (Public Review):

This is a manuscript that attempts to validate *Plasmodium* M1 alanyl aminopeptidase as a target for antimalarial drug development. The authors provide evidence that MIPS2673 inhibits recombinant enzymes from both Pf and Pv and is selective over other proteases. There is in vitro antimalarial activity. Chemoproteomic experiments demonstrate selective targeting of the PfA-M1 protease.

This is a continuation of previous work focused on designing inhibitors for aminopeptidases by a subset of these authors. Medicinal chemistry explorations resulted in the synthesis of MIPS2673 which has improved properties including potent inhibition of PfA-M1 and PvA-M1 with selectivity over a closed related peptidase. The compound also demonstrated selectivity over several human aminopeptidases and was not toxic to HEK293 cells at 40 uM. The activity against *P. falciparum* blood-stage parasites was about 300 nM.

Thermal stability studies confirmed that PfA-M1 was a binding target, however, there were other proteins consistently identified in the thermal stability studies. This raises the question as to their potential role as additional targets of this inhibitor. The authors dismiss these because they are not metalloproteases, but further analysis is warranted. This is particularly important as the authors were not able to generate mutants using in vitro evolution of resistance strategies. This often indicates that the inhibitor has more than one target.

The next set of experiments focused on a limited proteolysis approach. Again several proteins were identified as interacting with MIPS2673 including metalloproteases. The authors go on to analyze the LiP-MS data to identify the peptide from PfA-M1 which putatively interacts with MIPS2673. The authors are clearly focused on PfA-M1 as the target, but a further analysis of the other proteins identified by this method would be warranted and would provide evidence to either support or refute the authors' conclusions.

The final set of experiments was an untargeted metabolomics analysis. They identified 97 peptides as significantly dysregulated after MIPS2673 treatment of infected cells and most of these peptides were derived from one of the hemoglobin chains. The accumulation of peptides was consistent with a block in hemoglobin digestion. This experiment does reveal a potential functional confirmation, but questions remain as to specificity.

Overall, this is an interesting series of experiments that have identified a putative inhibitor of PfA-M1 and PvA-M1. The work would be significantly strengthened by structure-aided analysis. It is unclear why putative binding sites cannot be analyzed via specific mutagenesis of the recombinant enzyme. In the thermal stability and LiP-MS analysis, other proteins were consistently identified in addition to PfA-M1 and yet no additional analysis was undertaken to explore these as potential targets. The metabolomics experiments were potentially interesting, but without significant additional work including different lengths of treatment and different stages of the parasite, the conclusions drawn are overstated. Many treatments disrupt hemoglobin digestion - either directly or indirectly and from the data presented here it is premature to conclude that treatment with MIPS2673 directly inhibits hemoglobin digestion. Finally, the potency of this compound on parasites grown in vitro is 300 nM - this would need improvements in potency and demonstration of in vivo efficacy in the SCID mouse model to consider this a candidate for a drug.

Summary:

Overall, this is an interesting series of experiments that have identified a putative inhibitor of the Plasmodium M1 alanyl aminopeptidases, PfA-M1 and PvA-M1.

Strengths:

The main strengths include the synthesis of MIPS2673 which is selectively active against the enzymes and in whole-cell assay.

Weaknesses:

The weaknesses include the lack of additional analysis of additional targets identified in the chemoproteomic approaches.

<https://doi.org/10.7554/eLife.92990.1.sa0>